Working paper | Risk and capital research

A Coupled Power–Thermal–Cyber Framework for the Actuarial Pricing and Insurance of Hyperscale Data Centers

An integrated framework joining electrical reliability engineering, mathematical statistics of extremes, classical risk theory, and the microeconomic and macroeconomic consequences of insurance pricing.

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Draft | May 18, 2026
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Abstract

The data center asset class has, within a single decade, evolved from specialised commercial real estate into systemically critical digital infrastructure. Individual hyperscale campuses now carry total insurable values between \$10 bn and \$30 bn, yet the global property catastrophe market struggles to assemble more than \$5 bn of single-asset capacity. The combination of (i) unprecedented concentration of capital, (ii) novel physical failure modes (lithium-ion thermal runaway, immersion-cooling water-electrical contact, accelerated cooling failure under AI training loads), (iii) deterministic Service-Level-Agreement liabilities cascading into contingent business interruption, and (iv) tightening cyber–physical coupling, has produced a risk that does not fit any traditional property/casualty pricing template.

We propose an integrated actuarial framework that joins three historically separated disciplines: *electrical reliability engineering*, *mathematical statistics of extremes*, and *classical risk theory*. Our main contributions are:

1. A continuous-time **coupled state-space model** $\mathbf{X}_t = (V_t, T_t, C_t)$ of bus voltage, cold-aisle temperature, and cyber load, with an Arrhenius-accelerated instantaneous failure hazard $\lambda_t = h(\mathbf{X}_t)$; thermal runaway is treated as a first-hitting time.
2. A **mapping from Uptime Tier engineering availability to actuarial claim frequency**, allowing underwriters to translate $N + 1/2N$ topology directly into Poisson intensity.
3. A **compound severity model** blending a lognormal body with a generalised-Pareto tail fitted via Peaks-over-Threshold, recognising the bimodal nature of small operational losses and rare catastrophic site losses.
4. A **Gaussian–copula coupling** between the cyber intensity process and the thermal/electrical degradation process, capturing tail dependence that an independence assumption systematically under-prices.
5. A **hybrid indemnity–parametric product design** in which SCADA telemetry triggers fast partial payments and an indemnity layer settles residual basis risk.

We close with a worked pricing example for a 200 MW critical-IT-load hyperscale facility, including risk-loaded technical premium, Solvency II Standard Capital, and a per-risk excess-of-loss reinsurance illustration.

Keywords data center insurance; reliability engineering; thermal runaway; compound Poisson; Generalised Pareto; copulas; parametric trigger; SCADA; PML; Solvency II.

JEL / MSC G22, C46, 62P05, 90B25.

Contents

1. Introduction

1.1. The capacity problem

Hyperscale data center investment is projected by S&P Global Ratings to exceed \$300 bn per year by 2027. Single-asset Total Insured Values (TIV) routinely sit in the 10 to 30 bn USD range; the broking market estimates absorbable single-event loss capacity at no more than \$2 – 2.5 bn [3, 6]. Builders'-risk towers of \$2.7 bn (Marsh *Nimbus*) and lifecycle programmes of \$3.5 bn (Aon) have been launched specifically to syndicate this gap. The structural shortfall is large enough that Swiss Re estimates a single hyperscale site could generate close to \$10 bn of natural-catastrophe insured loss — $\sim 9\,\%$ of global insured catastrophe losses in a typical year. Aggregation is acute: most hyperscale campuses are clustered in a handful of ZIP codes (Northern Virginia, Phoenix, Dublin, Singapore).

1.2. Why traditional models fail

Conventional property pricing decomposes risk into a few well-trodden sub-perils (fire, water, named storm, machinery breakdown) and treats business interruption as a deterministic function of property damage. For a data center this decomposition is inadequate for three reasons.

(R1) Multi-physics coupling. A small electrical fault upstream of an Uninterruptible Power Supply (UPS) can cascade into a lithium-ion thermal runaway, then propagate through fire to mechanical cooling plant, and finally to total IT load loss. The recent NIRS Daejeon fire (26 Sept 2025), triggered by a single NMC battery cell during a relocation, brought down 647 Korean government services for $\approx$ 22 hours [7]. None of the conventional univariate fire/electrical sub-models capture the coupling explicitly.

(R2) Decoupled BI and downstream SLA liability. A *successful* cyber-defence lockdown can still cause an outage and trigger multi-million-dollar SLA penalties to colocation tenants. The indemnity-based BI mechanism, which compensates the operator's own income loss, does not address the contractual liability layer.

(R3) Concentration and operational tail dependence. Hyperscale tenants concentrate at a handful of Availability Zones. The AWS US-East-1 outage of 20 October 2025 caused \$38 M – \$581 M of insurable loss [5] simply by disturbing one regional service. This is the digital analogue of an earthquake zone — requiring catastrophe modelling, not standard fire underwriting.

1.3. Our contribution

Building on the early parametric work of Parametrix [9] on cloud-outage indices, we develop a fully integrated technical framework usable by actuaries, reinsurers, and electrical engineers in a common notation. Section 2 introduces the empirical risk taxonomy and anchors. Sections 3 and 4 present the electrical topology together with its reliability mathematics, including a complete fault-tree-to-Markov mapping. Section 5 develops the coupled state-space hazard model. Sections 6–8 treat frequency, severity, and cyber–physical dependence. The pricing, product design, and capital framework follow in Sections 9–12, before a fully worked example in Section 11.

2. Empirical anchors and risk taxonomy

2.1. The Uptime Tier system as a prior

The Uptime Institute Tier Classification, introduced in 1995 and now issued in 122 countries, is the de facto operational reliability prior for the asset class. It is performance-based rather than prescriptive, but its expected annual availabilities are widely cited (Table 1).

Table 1: Uptime Institute Tier classification and implied availability.

Tier	Topology	Availability A	Downtime (h/yr)	Electrical & cooling
I	N	99.671%	28.8	Single path, no redundancy
II	$N+1$	99.741%	22.0	Redundant components, single path
III	$N+1$ concurrent maintainable	99.982%	1.6	Multiple paths, one active
IV	$2N$ or $2N+1$ fault-tolerant	99.995%	0.4	Multiple active paths

We will use these availabilities as the engineering prior on annual *site-down hours*, and derive the actuarial claim frequency λ by an inverse mean-failure-time mapping (see §4.3).

2.2. The empirical peril mix

The Uptime Institute annual outage analysis attributes recent public outages roughly as follows: connectivity / network 29%, power systems 21%, cooling 12%, software/configuration 12%, on-site fire (often UPS-battery related) 7%, with the remainder spread across human factors, third-party providers and other causes. The fire share is disproportionately destructive: with the migration to lithium-ion UPS chemistries (now ≈ 38.5 % of the data center battery market versus 15 % in 2020 [8]), the mean severity of a fire-induced outage has risen sharply.

2.3. Loss components

For a single site i and a single insured event, total economic loss $L^{(i)}$ admits the decomposition

$$L^{(i)} = \underbrace{L^{(i)}_{\mathrm{PD}}}_{\text{property damage}} + \underbrace{L^{(i)}_{\mathrm{BI}}}_{\text{operator BI}} + \underbrace{L^{(i)}_{\mathrm{SLA}}}_{\text{SLA liability}} + \underbrace{L^{(i)}_{\mathrm{CY}}}_{\text{cyber/data}} + \underbrace{L^{(i)}_{\mathrm{LIAB}}}_{\text{third-party}} . \tag{1}$$

Each sub-component has its own distributional regime and its own dependence structure. The classical assumption $L_{\mathrm{BI}} \propto L_{\mathrm{PD}}$ is empirically violated: cyber lockdowns and SLA breaches can occur without any physical damage at all.

3. Electrical topology of a Tier-IV hyperscale site

Pricing requires the same electrical situational awareness an electrical engineer carries when commissioning the site. We summarise the canonical $2N$ topology — the dominant architecture for new hyperscale AI campuses — in the single-line wireframe of Figure 1.

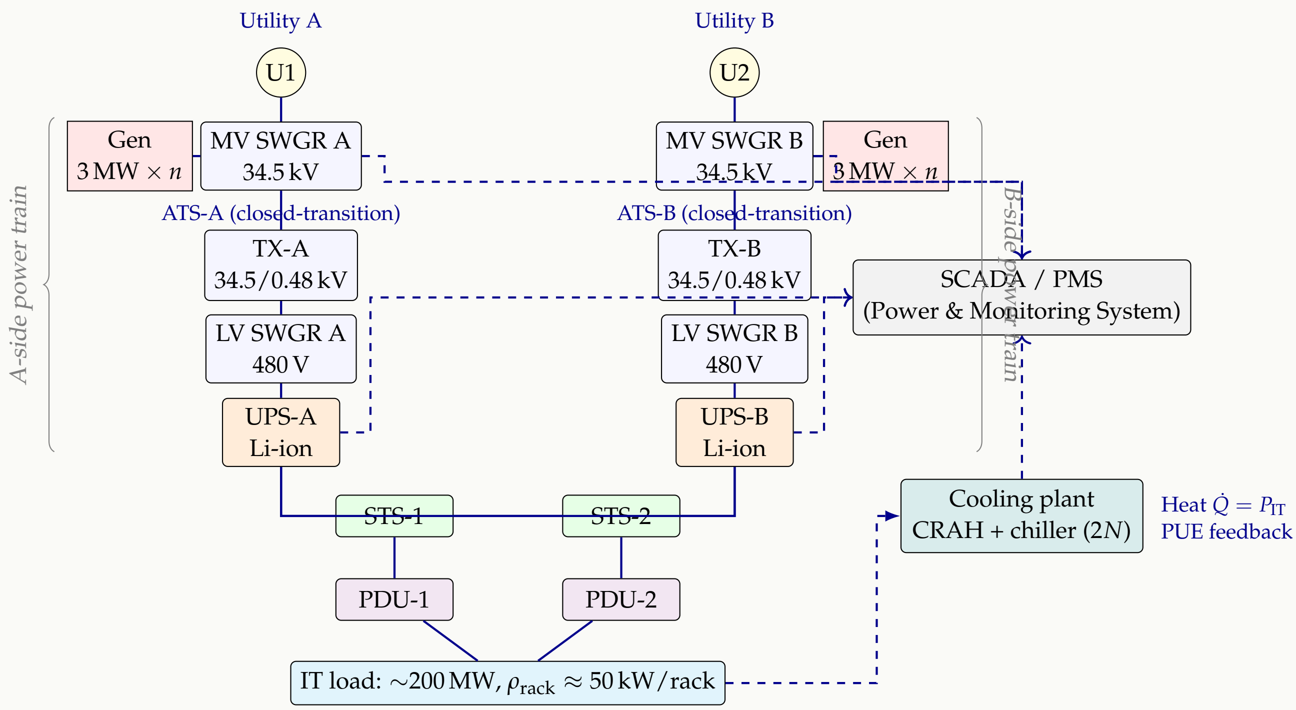

Figure 1: Single-line wireframe of a Tier-IV $2N$ hyperscale data center. Solid lines are power feeders; dashed lines are SCADA telemetry. Each half (A, B) carries 100 % of the design IT load, so any single component on one side may fail or be taken out for maintenance with no load impact. The dual independent utility feeds, dual generator plants, dual UPS rooms and dual cooling trains are the physical expression of the engineering $2N$ availability target of 99.995%.

3.1. Engineering parameters that enter pricing

The underwriting-relevant electrical parameters are summarised in Table 2. They will reappear in the hazard rate specification of §5.

Table 2: Underwriting-relevant electrical engineering parameters.

Symbol	Quantity	Typical value (Tier IV, AI campus)
P_{IT}	critical IT load	50 MW–500 MW
PUE	power usage effectiveness	1.10–1.45 (best-in-class 1.04)
ρ_{rack}	rack power density	30 kW/rack to 120 kW/rack
T_{ca}	cold-aisle setpoint	18 °C to 27 °C (ASHRAE A1)
η_{UPS}	UPS double-conversion efficiency	0.96–0.98
τ_{auto}	generator autonomy on diesel	24–72 h
E_{batt}	UPS battery energy	5 min to 15 min of full load
N_{cells}	lithium-ion cells per UPS room	10^4–10^5

4. Reliability mathematics: from components to the site

We assume the reader is familiar with the basic identity $R(t) = \exp(-\int_0^t \lambda(s)\mathrm{d}s)$ and the relationship between failure rate λ, MTBF, MTTR and steady-state availability

$$A = \frac{\mathrm{MTBF}}{\mathrm{MTBF} + \mathrm{MTTR}}, \qquad \mathrm{MTBF} = \frac{1}{\lambda}, \qquad U = 1 - A. \tag{2}$$

4.1. Fault tree representation

Let T denote the top event "IT load lost" at a particular site, and let $\{B_j\}_{j=1}^m$ denote the elementary events (component failures, human errors, exogenous shocks). A fault tree encodes T as a Boolean function of $\{B_j\}$,

$$T = \phi(B_1, \dots, B_m). \tag{3}$$

For a $2N$ topology the simplified structure is

$$T \Leftrightarrow \underbrace{(E_A \wedge E_B)}_{\text{both power trains}} \vee \underbrace{(C_A \wedge C_B)}_{\text{both cooling trains}} \vee \underbrace{F_{\mathrm{cat}}}_{\text{catastrophic peril}} \vee \underbrace{Y_{\mathrm{cy}}}_{\text{cyber lockdown}}, \tag{4}$$

where each $E_\bullet$ is itself a series chain (utility OR generator OR ATS OR transformer OR UPS OR PDU). Figure 2 shows the expanded tree.

For *instantaneous* (steady-state) probabilities and *rare*, independent component failures,

$$\mathbb{P}(T) \approx \mathbb{P}(E_A)\mathbb{P}(E_B) + \mathbb{P}(C_A)\mathbb{P}(C_B) + \mathbb{P}(F_{\mathrm{cat}}) + \mathbb{P}(Y_{\mathrm{cy}}), \tag{5}$$

where each $\mathbb{P}(E_\bullet) = 1 - \prod_k (1 - q_k)$ for the relevant series chain of unavailabilities q_k. The independence assumption is abandoned in §8.

4.2. Markov chain of plant states

A more dynamic representation is the continuous-time Markov chain $\{Z_t\}$ on

$$\mathcal{S} = \{\mathrm{OK},\ \mathrm{Deg}_1,\ \mathrm{Deg}_2,\ \mathrm{F}\}, \tag{6}$$

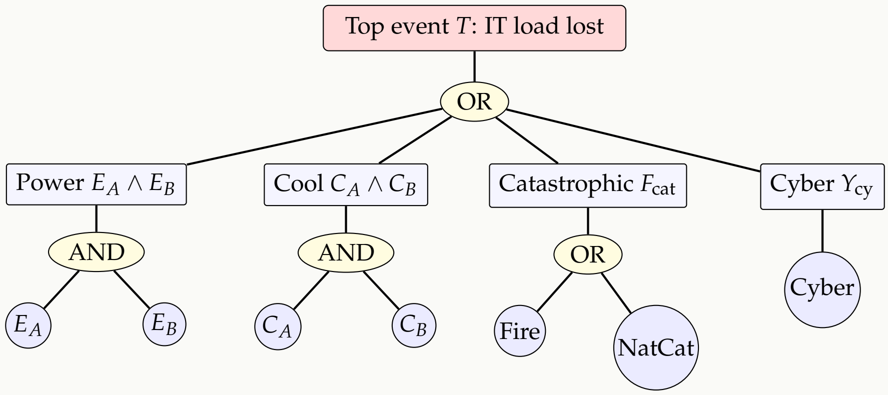

Figure 2: Simplified fault tree for the top event T = “IT load lost.” Each $E_\bullet$ is itself a series chain of elementary component events (utility, generator, ATS, transformer, switchgear, UPS, STS, PDU); these are not expanded here for clarity. The four top-level branches are not independent — a fire in the UPS room can simultaneously fail E_A and trigger fire suppression that shuts cooling.

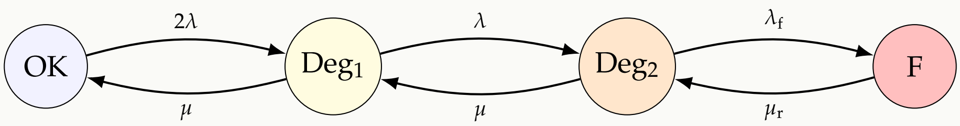

Figure 3: Continuous-time Markov chain on plant states for a $2N$ data center. λ: per-train degradation rate. μ: per-train repair rate. λ_{f}: degraded-to-failed rate, which is dramatically higher than λ because the system has lost fault tolerance. μ_{r}: full-site restoration rate.

corresponding to (both trains healthy, one train down, both trains down on one subsystem, total failure). For a $2N$ site:

The steady-state distribution $\boldsymbol{\pi} = (\pi_0, \pi_1, \pi_2, \pi_F)$ solves $\boldsymbol{\pi}\,\mathbf{Q} = \mathbf{0}$, $\boldsymbol{\pi}\mathbf{1} = 1$, where the generator is

$$\mathbf{Q} = \begin{pmatrix} -2\lambda & 2\lambda & 0 & 0 \\ \mu & -(\mu+\lambda) & \lambda & 0 \\ 0 & \mu & -(\mu+\lambda_{\text{f}}) & \lambda_{\text{f}} \\ 0 & 0 & \mu_{\text{r}} & -\mu_{\text{r}} \end{pmatrix}. \tag{7}$$

The unavailability $U = \pi_F$ admits a closed-form expression that we can equate to $1 - A$ from Table 1; this calibration identifies the per-component λ once μ and μ_{r} are fixed by site operating procedures.

Proposition 4.1. *For the chain in Figure 3 with $\lambda \ll \mu$, the steady-state unavailability satisfies, to leading order,*

$$U \approx \frac{2\lambda^2\lambda_{\text{f}}}{\mu^2\mu_{\text{r}}}. \tag{8}$$

Sketch. With $\lambda \ll \mu$, the chain is heavily attracted to OK. Standard results from Trivedi [11] give $\pi_1 \approx 2\lambda/\mu$, $\pi_2 \approx 2\lambda^2/\mu^2$, $\pi_F \approx 2\lambda^2\lambda_{\text{f}}/(\mu^2\mu_{\text{r}})$. □

Equation (8) is the bridge from electrical engineering specifications to actuarial frequency: ageing of any component drives λ up multiplicatively, and the cube of those rates appears in U. The framework is exquisitely sensitive to component reliability — exactly the underwriting result needed.

4.3. From availability to claim frequency

If we declare a "site outage event" to be any visit by $\{Z_t\}$ to F, the number of such visits in $[0,1]$ year is, in steady state, well-approximated by a Poisson process with intensity

$$\lambda^{\text{out}} \;=\; \pi_2\,\lambda_{\mathrm{f}} \;\approx\; \frac{2\lambda^2\lambda_{\mathrm{f}}}{\mu^2}\ \text{per year.} \tag{9}$$

This is our *engineering-derived prior* on actuarial frequency. We will see in §6 how to update it from empirical claims data.

5. The coupled power–thermal–cyber failure model

5.1. State space and dynamics

The novelty of our framework is to treat the data center as a *coupled multi-physics system* rather than as a collection of independent components. Let

$$\mathbf{X}_t \;=\; \big(V_t,\ T_t,\ C_t\big)^{\top} \in \mathbb{R}^3, \tag{10}$$

where

- V_t: the prevailing per-unit bus voltage at the main LV switchgear ($V_t \equiv 1$ in normal operation),
- T_t: the cold-aisle return temperature deviation above setpoint (K),
- C_t: an abstract cyber-load process (e.g. rate of intrusion attempts per second, log-transformed).

We posit the following stochastic differential equations:

$$\boxed{\begin{aligned} \mathrm{d}V_t &= -\kappa_V\big(V_t-1\big)\,\mathrm{d}t + \sigma_V\,\mathrm{d}W_t^V - J_t^V\,\mathrm{d}N_t^V,\\ \mathrm{d}T_t &= \alpha\,\big[g(V_t)\,P_{\mathrm{IT}} - q(\mathrm{PUE})\,c(V_t)\big]\mathrm{d}t + \sigma_T\,\mathrm{d}W_t^T,\\ \mathrm{d}C_t &= \kappa_C\big(\bar{c}-C_t\big)\mathrm{d}t + \sigma_C\,\mathrm{d}W_t^C, \end{aligned}} \tag{11}$$

with independent standard Brownian motions (W^V, W^T, W^C) subject to the correlation $\mathrm{d}W^V\mathrm{d}W^T = \rho_{VT}\mathrm{d}t$, $\mathrm{d}W^T\mathrm{d}W^C = \rho_{TC}\mathrm{d}t$, $\mathrm{d}W^V\mathrm{d}W^C = \rho_{VC}\mathrm{d}t$, all initialised at $\mathbf{X}_0 = (1, 0, \bar{c})$, and where N_t^V is a Poisson process counting upstream voltage-sag events with intensity β_V.

Reading the model.

- The voltage equation is an Ornstein–Uhlenbeck process pulled back to nominal ($V = 1$), perturbed by Gaussian small-signal noise $\sigma_V \mathrm{d}W^V$ and discrete sag-jumps J_t^V of random size at rate β_V. Real SCADA voltage traces are well-modelled in this form.
- In the thermal equation, the function $g(v) = \mathbb{1}_{\{v>v_{\min}\}}$ acts as a binary indicator that IT continues to draw rated power above the UPS dropout threshold $v_{\min}$, while $c(v)$ is the cooling-system capacity which collapses below the same threshold (cooling plant pulled off-grid). The parameter $q(\mathrm{PUE}) = (\mathrm{PUE}-1)\,P_{\mathrm{IT}}$ converts the facility overhead into reject heat. α is the lumped thermal capacitance of the white space.
- The cyber-load equation is again Ornstein–Uhlenbeck, capturing the noisy background of intrusion attempts. Spikes are modelled via a separately-counted compound Poisson process below.

5.2. Coupled instantaneous failure hazard

The instantaneous hazard rate that the IT critical load is lost in the next time-step combines three accelerative effects:

$$\boxed{\lambda_t \;=\; \underbrace{\lambda_0\, e^{-E_a/k_B T_t}}_{\text{Arrhenius thermal}} \cdot \underbrace{\Big(\tfrac{V_t}{V_{\text{nom}}}\Big)^{-\gamma_V}}_{\text{voltage-stress}} \cdot \underbrace{\exp(\beta_C\, C_t)}_{\text{cyber-stress}}.} \tag{12}$$

The Arrhenius factor comes from the standard semiconductor reliability acceleration model: E_a is an effective activation energy (typically 0.6 – 0.9 eV for IC failures), k_B is Boltzmann's constant. A $\Delta T = 10\,\text{K}$ rise above setpoint roughly halves component MTBF [2], which is the headline industrial "10 °C rule."

The voltage-stress factor reflects accelerated insulation ageing with $\gamma_V \in [3, 7]$ for liquid-filled transformers under the Arrhenius–inverse-power-law (AIPL) model.

The cyber-stress factor is the cleanest novelty: an Esscher-style exponential tilt of the baseline rate by the prevailing cyber-load.

5.3. Catastrophic transitions as first-hitting times

A subset of catastrophic events are best modelled not as smooth hazard increases but as the first time a state variable crosses a critical boundary.

Definition 5.1 (*Thermal runaway hitting time*). *Define the thermal runaway barrier $T^* = T_{\text{runaway}} - T_{\text{setpoint}}$, e.g. $T^* \approx 40\,\text{K}$ for Li-ion NMC chemistry. The thermal runaway hitting time is*

$$\tau^* \;=\; \inf\{\, t > 0 : T_t \geq T^* \,\}. \tag{13}$$

For the thermal SDE in (11) the distribution of τ^* solves the Kolmogorov backward equation; under mean-reverting drift it is well-approximated by an inverse-Gaussian distribution [4]:

$$\tau^* \sim \mathrm{IG}(\mu_\tau, \lambda_\tau), \qquad f_{\tau^*}(t) \;=\; \sqrt{\tfrac{\lambda_\tau}{2\pi t^3}}\, \exp\Big\{-\tfrac{\lambda_\tau (t-\mu_\tau)^2}{2\mu_\tau^2 t}\Big\}, \tag{14}$$

with mean $\mu_\tau = T^*/\alpha\, \mathbb{E}[\text{net heat in}]$ and shape λ_τ identified from σ_T and the boundary distance.

5.4. Deterministic limit: the operator's grace period

The stochastic first-hitting characterisation has a clean deterministic counterpart that engineers reach for first and that operators actually use in pre-incident drills. Setting $\sigma_T \to 0$ and $V_t \equiv 1$ in (11) reduces the thermal equation to the lumped-capacitance heat balance

$$m\, c_p \, \frac{\mathrm{d}T(t)}{\mathrm{d}t} \;=\; Q_{\text{IT}} \;-\; UA\,\big(T(t) - T_{\text{amb}}\big), \tag{15}$$

where m (kg) and c_p (J kg^{-1}K^{-1}) are the lumped mass and specific heat of air and structure in the white space, UA (W K^{-1}) is the lumped envelope conductance, $Q_{\text{IT}} = P_{\text{IT}}$ is the IT reject heat and T_{amb} is the ambient outdoor temperature. Equation (15) is a linear first-order ODE with closed-form solution

$$T(t) \;=\; \underbrace{T_{\text{amb}} + \tfrac{Q_{\text{IT}}}{UA}}_{T_\infty} \;+\; \Big(T_0 - T_{\text{amb}} - \tfrac{Q_{\text{IT}}}{UA}\Big)\, \exp\Big(-\tfrac{UA}{mc_p}\, t\Big). \tag{16}$$

Note that for a typical hyperscale hall in cooling-failure mode, Q_{IT}/UA is far above any safe room temperature: cooling *must* resume before the asymptote T_∞ is approached. Solving (16) for the first time T reaches the ASHRAE A1 operating limit T_{crit} (32 °C for general use) gives the **deterministic grace period**

$$\boxed{t_{\text{crit}} \;=\; -\,\frac{mc_p}{UA}\,\ln\!\left(\frac{T_{\text{crit}} - T_{\text{amb}} - Q_{\text{IT}}/UA}{T_0 - T_{\text{amb}} - Q_{\text{IT}}/UA}\right).} \tag{17}$$

The stochastic counterpart τ^* from (13)–(14) is well-approximated by a distribution centred at t_{crit} with dispersion driven by σ_T.

Remark

For a typical 20 MW IT hall ($Q_{\text{IT}} = 20\,\text{MW}$), $mc_p \approx 3.5 \times 10^6\,\text{J/K}$, $UA \approx 10^4\,\text{W/K}$, $T_0 = 22\,°\text{C}$, $T_{\text{amb}} = 30\,°\text{C}$, $T_{\text{crit}} = 32\,°\text{C}$, (17) gives $t_{\text{crit}} \approx 3.5$ minutes. This is the engineering reality that drives the 24/7 rotation of redundant chillers, and the actuarial reality that even Tier-IV sites with 15 minute battery autonomy are running closer to the cooling boundary than to the power boundary.

Novel contribution

Why this matters. Treating thermal runaway as a first-hitting time means the protective effect of *off-gas detection* (which shifts the boundary T^* effectively to a lower temperature T^*_{eff} because mitigation can be applied earlier) enters the model *multiplicatively in the hazard*, not additively in the loss. This explains why off-gas detection retrofits have been observed to move site-level loss costs by 40–60 %, far more than a naive reduction-in-frequency calculation predicts: the lengthened tail of τ^* removes precisely the most extreme right tail of the loss distribution.

6. Frequency modelling

6.1. Annual count of insurable outages

Let N_i be the number of outage events at site i in a policy year. The naïve actuarial benchmark is the homogeneous Poisson

$$N_i \sim \text{Poisson}(\lambda_i), \qquad \mathbb{E}[N_i] = \text{Var}(N_i) = \lambda_i. \tag{18}$$

Empirical experience with multi-site portfolios overwhelmingly shows over-dispersion ($\text{Var}/\mathbb{E} > 1$), driven by site heterogeneity in operational maturity, climate exposure and equipment age. We therefore adopt a **Poisson–Gamma frailty** mixed model:

$$N_i \,|\, \Theta_i \sim \text{Poisson}(\lambda_i \Theta_i), \qquad \Theta_i \overset{\text{iid}}{\sim} \text{Gamma}(\nu, \nu), \tag{19}$$

which marginally yields a Negative Binomial $N_i \sim \text{NB}\big(\nu,\, \nu/(\nu + \lambda_i)\big)$ with

$$\mathbb{E}[N_i] = \lambda_i, \qquad \text{Var}(N_i) = \lambda_i(1 + \lambda_i/\nu). \tag{20}$$

6.2. The frequency regression

The site-specific rate λ_i is parameterised log-linearly in underwriting covariates:

$$\log \lambda_i \;=\; \beta_0 + \beta_1 \log P_{\text{IT},i} + \beta_2\,(\text{PUE}_i - 1) + \beta_3\,\text{Age}_i + \beta_4\,\mathbf{1}\{\text{Tier} < \text{IV}\}_i + \beta_5\,\text{ClimateScore}_i. \tag{21}$$

The coefficient on PUE deserves attention: *higher PUE means more of the energy budget is lost to mechanical and electrical overhead*. Empirically this co-varies with cooling stress and with lower-quality equipment. We expect $\beta_2 > 0$. The IT-load coefficient absorbs the simple intuition that bigger sites have more components and therefore more failure opportunities, but *economies of redundancy* typically push $\beta_1 < 1$.

6.3. Engineering prior, statistical posterior

Equation (9) gives the engineering prior $\lambda_i^{\text{out,eng}}$ from the Tier-class availability. In a Bayesian setting, we treat it as the prior mean and update with claims experience $\mathbf{D}_i = \{n_{i,t}, t = 1, \ldots, T\}$:

$$\pi(\lambda_i \,|\, \mathbf{D}_i) \;\propto\; \pi(\lambda_i) \prod_{t=1}^{T} \frac{(\lambda_i)^{n_{i,t}} e^{-\lambda_i}}{n_{i,t}!}. \tag{22}$$

If the prior is Gamma, this is conjugate and gives a closed-form posterior. This is precisely the *credibility* framework of Bühlmann–Straub, with the engineering prior playing the role of the collateral information.

REMARK

Cox/doubly-stochastic structure. The Poisson–Gamma frailty in (19) is precisely a *doubly stochastic* (Cox) Poisson process: the intensity $\lambda_i \Theta_i$ is itself random, drawn at the site level. If we let the intensity depend on time-varying covariates such as $T_{\text{amb}}(t)$, the model generalises naturally to

$$\lambda_i(t) \;=\; \lambda_i \cdot \exp\big(\gamma_1 \max(0, T_{\text{amb}}(t) - T_{\text{design}}) + \gamma_2 \mathbf{1}_{\{Z_t = \text{Deg}_2\}}\big),$$

which couples the actuarial frequency directly to the Markov chain state of §4.2 and to a meteorological covariate. The unconditional claim count is then a Cox process whose intensity inherits the dynamics of both the plant and the weather.

7. Severity modelling and the tail

7.1. Compound aggregate loss

Total annual loss for site i is the compound random variable

$$S_i \;=\; \sum_{j=1}^{N_i} X_{i,j}, \qquad X_{i,j} \overset{\text{iid}}{\sim} F_X. \tag{23}$$

Both the mean and variance of S_i are given by Wald's identities:

$$\mathbb{E}[S_i] = \mathbb{E}[N_i]\,\mathbb{E}[X], \tag{24}$$

$$\operatorname{Var}(S_i) = \mathbb{E}[N_i]\ \operatorname{Var}(X) + \operatorname{Var}(N_i)\,\mathbb{E}[X]^2. \tag{25}$$

For the NB frequency assumption, the second identity becomes

$$\operatorname{Var}(S_i) \;=\; \lambda_i\,\mathbb{E}[X^2] + \frac{\lambda_i^2}{\nu}\,\mathbb{E}[X]^2, \tag{26}$$

making explicit how frailty inflates the variance over the Poisson case.

7.2. Body–tail severity decomposition

The severity distribution F_X has a markedly bimodal character:

- **Attritional losses** (PDU failures, single rack burns, short cooling excursions) cluster around \$100k–\$5M and fit a lognormal body very well.
- **Catastrophic losses** (full-site fire, total power train loss, lithium-ion cascading event) are heavy-tailed and follow a power law.

We adopt a *Peaks-over-Threshold* (POT) decomposition: choose a threshold u (e.g. the 95th percentile of historical claims), and model

$$F_X(x) \;=\; \begin{cases} F_{\text{body}}(x), & x < u, \\ 1-(1-F_X(u))\left[1+\xi\dfrac{x-u}{\sigma}\right]^{-1/\xi}, & x \geq u. \end{cases} \tag{27}$$

The tail is a Generalised Pareto with shape ξ and scale σ, justified by the Pickands–Balkema–de Haan theorem [1, 10]: for any distribution in the maximum domain of attraction of an extreme value distribution, exceedances above a high enough threshold converge to a GPD. Maximum-likelihood estimates of ξ for data-center catastrophic losses are typically in the $\xi \in (0.2, 0.6)$ range — meaningfully heavy-tailed.

7.3. Probable Maximum Loss and Maximum Foreseeable Loss

Lenders and reinsurers reference two engineering loss metrics adapted to data centers:

$$\text{PML}_\alpha \;=\; \text{VaR}_\alpha(X \mid X > u) \;=\; u + \frac{\sigma}{\xi}\left[\left(\tfrac{1-\alpha}{p_u}\right)^{-\xi} - 1\right], \tag{28}$$

$$\text{MFL} \;=\; \text{deterministic worst credible loss assuming all controls fail.} \tag{29}$$

For a $\xi = 0.4$, $\sigma = \$50M$, $u = \$10M$ tail and $\alpha = 99.5\%$ with $p_u = 0.05$, the PML works out to about \$870M — consistent with broker estimates for a 100 MW Tier-III site.

7.4. Operationalising BI severity via SLA structure

The pure-statistical severity above tells us *how big* a loss is but says nothing about *why*. For data center operators, the loss decomposition (1) contains an item $L_{\text{BI}}^{(i)}$ whose magnitude is governed by an SLA contract between the operator and its tenants. A single outage of length τ generates a piecewise BI loss

$$\boxed{X^{\text{BI}}(\tau) \;=\; C_{\text{SLA}}\,\max(0,\,\tau-\tau_0) \;+\; \kappa\,\mathbf{1}\{\tau > t_{\text{crit}}\} \;+\; \rho_{\text{churn}}\,\tau^2\,,} \tag{30}$$

with the interpretation

- τ_0: the SLA-permitted deductible window (e.g. "five nines" $\Rightarrow$ $\tau_0 \approx 5.26$ min/yr).
- C_{SLA}: contractual penalty rate per unit downtime above τ_0 (US\$/min). Most colocation contracts set C_{SLA} to one to three monthly recurring revenue (MRR) equivalents per cumulative downtime quarter, with caps.
- κ: a step penalty triggered the moment downtime exceeds the thermodynamic grace period t_{crit} derived in (17); this is where "customers start migrating" in real-world incident accounts.
- $\rho_{\text{churn}}\,\tau^2$: a quadratic churn term capturing the empirical fact that the marginal customer-loss rate grows with downtime, not stays flat.

The BI duration τ itself is well-modelled as lognormal, $\ln\tau \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2)$, calibrated from MTTR experience. Combining with (30) and integrating out τ gives a closed-form $\mathbb{E}[X^{\text{BI}}]$ in terms of the lognormal CDF Φ, useful for fast quotation:

$$\begin{aligned} \mathbb{E}[X^{\text{BI}}] \;=\; & C_{\text{SLA}}\left[\,e^{\mu_\tau+\sigma_\tau^2/2}\bar{\Phi}(z_0) - \tau_0\,\bar{\Phi}(z_0')\,\right] \\ & +\; \kappa\,\bar{\Phi}(z_{\text{crit}}') \;+\; \rho_{\text{churn}}\,e^{2\mu_\tau+2\sigma_\tau^2}, \end{aligned} \tag{31}$$

with $z_0 = (\ln \tau_0 - \mu_\tau - \sigma_\tau^2)/\sigma_\tau$, primed versions at $\sigma_\tau^2 = 0$, and $\bar{\Phi} = 1 - \Phi$. The actuarial contribution of the κ term is highly non-linear in t_{crit}, which links underwriting directly to the engineering parameters mc_p and UA from (17).

8. Cyber–physical dependence via copula

8.1. Why independence fails

Cyber incidents and physical failures are not independent. Three empirical mechanisms link them: (i) a successful intrusion of the Building Management System can deliberately mis-set cooling controls; (ii) physical events (a fire, a flood) often disable cyber-defence appliances along with everything else; (iii) major cyber-events and major physical events both correlate with operational human-factor stress (peak utilisation, contractor activity).

8.2. Gaussian copula linkage

Let S_i^{phy} and S_i^{cy} be the marginal annual loss aggregates from physical and cyber peril streams, with marginal CDFs F^{phy} and F^{cy}. We link them via a bivariate Gaussian copula:

$$C_\rho(u_1, u_2) \;=\; \Phi_2\big(\Phi^{-1}(u_1), \Phi^{-1}(u_2); \rho\big), \tag{32}$$

so that the joint distribution is

$$F(s_1, s_2) \;=\; C_\rho\big(F^{\text{phy}}(s_1),\, F^{\text{cy}}(s_2)\big). \tag{33}$$

The total site loss is then $S_i = S_i^{\text{phy}} + S_i^{\text{cy}}$, with variance

$$\operatorname{Var}(S_i) \;=\; \operatorname{Var}(S_i^{\text{phy}}) + \operatorname{Var}(S_i^{\text{cy}}) + 2\rho\sqrt{\operatorname{Var}(S_i^{\text{phy}})\operatorname{Var}(S_i^{\text{cy}})}. \tag{34}$$

> **REMARK**
>
> The Gaussian copula is a starting point but is asymptotically tail *independent*, which means it understates joint extremes. For the tail layer of a reinsurance treaty, a t-copula with low degrees of freedom (e.g. $\nu = 4$) or a Gumbel copula is preferable. The calibration data are scarce; expert elicitation of $\tau_{\text{Kendall}} = 0.15$–$0.25$ is a reasonable industry prior.

8.3. Gumbel copula for the tail layer

The Gaussian-copula tail-independence defect is repaired with an Archimedean Gumbel copula

$$C_\theta(u_1, u_2) \;=\; \exp\Big\{-\big[(-\ln u_1)^\theta + (-\ln u_2)^\theta\big]^{1/\theta}\Big\}, \qquad \theta \ge 1, \tag{35}$$

where $\theta = 1$ recovers independence and $\theta \to \infty$ yields perfect comonotonicity. The Gumbel admits asymptotic upper tail dependence

$$\lambda_U \;=\; \lim_{u \uparrow 1} \mathbb{P}\big(U_2 > u \,|\, U_1 > u\big) \;=\; 2 - 2^{1/\theta}, \tag{36}$$

so that $\theta = 1.5$ yields $\lambda_U \approx 0.41$ and $\theta = 3$ yields $\lambda_U \approx 0.74$. Kendall's τ is related to θ by the closed form $\tau_K = 1 - 1/\theta$, which is how the expert prior $\tau_K \in (0.15, 0.25)$ translates into $\theta \in (1.18, 1.33)$, hence $\lambda_U \in (0.20, 0.31)$ — a substantial joint-tail probability that the Gaussian copula assigns the value zero. The Gumbel is therefore our recommended choice for layers attaching at the 1-in-50 level and above, with the Gaussian retained for the attritional layer where the bulk of the data lives.

9. Pricing

9.1. Surplus process and the Cramér–Lundberg framing

Before computing premiums it is useful to fix the underlying surplus dynamics. For a portfolio absorbing the aggregate loss $S(t) = \sum_{j=1}^{N(t)} X_j$ at rate set by the Cox intensity of §6, the insurer's surplus evolves as

$$U(t) = u_0 + \Pi t - S(t), \tag{37}$$

where u_0 is initial capital and Π is the constant premium rate. The probability of ultimate ruin $\psi(u_0) = \mathbb{P}\big(\inf_{t\geq 0} U(t) < 0\big)$ is bounded above by Lundberg's inequality $\psi(u_0) \leq e^{-Ru_0}$ for the adjustment coefficient R solving $\mathbb{E}[e^{RX}] = 1 + \Pi R/\lambda$. In the data-center context, R is small (heavy-tailed X) and Lundberg's bound is loose, which is precisely why we move to the variance- and tail-based loading principles below rather than rely on classical ruin theory.

9.2. The pure premium

The pure (or risk) premium per site per year is

$$P_i^{\text{pure}} = \mathbb{E}[S_i] = \mathbb{E}[N_i] \cdot \mathbb{E}[X] = \lambda_i \, \mathbb{E}[X]. \tag{38}$$

For sites in a portfolio $\mathcal{I}$, the portfolio pure premium is $\sum_i P_i^{\text{pure}}$.

9.3. Risk loading principles

A pure premium does not compensate the insurer for taking on the volatility around $\mathbb{E}[S_i]$. The technical premium adds a *risk load* Π_{load}. We present four widely used principles and their interpretation:

Standard deviation principle.

$$P_i^{\text{SD}} = \mathbb{E}[S_i] + \theta \, \text{SD}(S_i), \qquad \theta > 0. \tag{39}$$

Simple, but does not distinguish upside from downside.

Variance principle.

$$P_i^{\text{V}} = \mathbb{E}[S_i] + a \, \text{Var}(S_i), \qquad a > 0, \tag{40}$$

which penalises portfolios with heavy tails more aggressively.

Esscher transform. For a parameter $h > 0$,

$$P_i^{\text{E}} = \frac{\mathbb{E}[\, S_i \, e^{hS_i}]}{\mathbb{E}[\, e^{hS_i}]}. \tag{41}$$

The Esscher principle has the appealing property of being additive on independent risks and corresponds to a change-of-measure construction naturally arising in financial mathematics.

Wang transform. The Wang transform [13] adjusts the survival function $\bar{F}_S(s) = 1 - F_S(s)$ via

$$\bar{F}_S^*(s) \;=\; \Phi\big(\Phi^{-1}(\bar{F}_S(s)) + \lambda^*\big), \tag{42}$$

and prices by $P_i^{\mathrm{W}} = \int_0^\infty \bar{F}_{S_i}^*(s)\mathrm{d}s$. The Wang transform produces the same prices as CAPM in the normal case and the same prices as Black–Scholes for log-normal payoffs — making it the natural choice for hybrid insurance/capital-markets structures (cat bonds, ILS) that are likely to absorb data center peak-layer risk over time.

9.4. Loading the data-center book

We propose a hybrid loading:

$$\boxed{P_i^{\mathrm{tech}} \;=\; \mathbb{E}[S_i] \;+\; \theta_1 \mathrm{SD}(S_i^{\mathrm{attr}}) \;+\; \theta_2\big[\mathrm{TVaR}_{0.995}(S_i^{\mathrm{cat}}) - \mathbb{E}[S_i^{\mathrm{cat}}]\big],} \tag{43}$$

splitting the attritional layer (low $\theta_1 \approx 0.3$ standard deviation load) from the catastrophic layer (high $\theta_2 \approx 0.15$ Tail-VaR-based load). This mirrors the actual reinsurance market's treatment.

10. Hybrid indemnity–parametric product

10.1. Trigger architecture

Pure indemnity products on data centers are slow to settle (complex forensics on equipment), inflate basis between sites, and bake in ex-post adversarial information asymmetry. Pure parametric products remove these frictions but introduce basis risk. We propose a **three-layer hybrid**:

Hybrid product

Layer 1 (parametric, fast). Triggered automatically by SCADA measurements of T_t, V_t exceeding pre-agreed thresholds for pre-agreed durations. Settled in days.

Layer 2 (parametric, cyber-trigger). Triggered by detection of cloud outage or measured downtime of named SaaS/PaaS/IaaS services exceeding a threshold, as already implemented by Parametrix [9].

Layer 3 (indemnity). Catastrophic property/BI/SLA layer for residual basis. Slower, larger limits, and reinsured via excess-of-loss treaties.

10.2. Basis risk decomposition

For Layer 1, basis risk is measured by

$$\mathrm{BR}_i \;=\; \mathbb{E}\Big[\,\big(L_i^{\mathrm{actual}} - L_i^{\mathrm{param}}\big)^2\Big], \tag{44}$$

which can be decomposed as

$$\mathrm{BR}_i \;=\; \underbrace{\mathrm{Var}\big(L_i^{\mathrm{actual}}\big)}_{\text{intrinsic}}\big(1 - R^2_{\mathrm{param,actual}}\big), \tag{45}$$

i.e. the proportion of loss variance not explained by the parametric trigger. Modern SCADA telemetry routinely achieves $R^2 > 0.85$ for power/cooling failures, but $R^2 < 0.4$ for cyber-induced losses. This explains why parametric *cyber* cover is harder to design than parametric *power* cover.

11. Worked example: pricing a 200 MW hyperscale site

Worked example

We price a fictional but representative 200 MW critical IT load, Tier-IV $2N$, AI-training-dominated hyperscale campus in Northern Virginia.

Inputs.

- Total Insured Value (TIV): \$8 bn.
- Engineering prior availability $A = 99.995\%$, hence implied site-outage rate from (8): $\lambda_{\text{eng}}^{\text{out}} \approx 0.05$ events / yr.
- NB frailty parameter $\nu = 3$ (moderate over-dispersion).
- Severity: lognormal body with $\mu_{\ln X} = 14.0$, $\sigma_{\ln X} = 1.4$ (*i.e.* median loss $\approx \$1.2M$, mean $\approx \$3.2M$).
- Tail: GPD with $\xi = 0.4$, $\sigma = \$60M$ above threshold $u = \$20M$; tail probability $p_u = 0.04$.
- Cyber–physical Kendall's $\tau = 0.20$, hence Gaussian copula $\rho = \sin(\pi\tau/2) \approx 0.31$.

Frequency. $\lambda_i \approx 0.05$. With NB($\nu = 3$), $\text{Var}(N_i) = 0.05(1 + 0.05/3) \approx 0.0508$. Annual probability of *any* outage is $1 - (1 + \lambda_i/\nu)^{-\nu} \approx 4.92\%$.

Severity moments. Body mean $\mathbb{E}[X^{\text{body}}] = e^{14.0+1.4^2/2} \approx \$3.2M$. Tail conditional mean $\mathbb{E}[X \mid X > u] = u + \sigma/(1-\xi) = \$20M + \$60M/0.6 = \$120M$. Unconditional mean $\mathbb{E}[X] = (1 - p_u)\mathbb{E}[X^{\text{body}}] + p_u\mathbb{E}[X^{\text{tail}}] \approx 0.96 \cdot \$3.2M + 0.04 \cdot \$120M \approx \$7.9M$.

Pure premium. $P_i^{\text{pure}} = \lambda_i \mathbb{E}[X] \approx 0.05 \cdot \$7.9M = \$395k$.

Risk-loaded technical premium. Standard-deviation load on attritional ($\theta_1 = 0.3$): \$118k. Tail load on catastrophic via $\text{TVaR}_{0.995} - \mathbb{E}$: with $\xi = 0.4$, $\sigma = \$60M$ the TVaR is \$315M, hence the per-site load ($\theta_2 = 0.15$) is $\approx \$240k$. **Technical premium** $\approx \$753k$ or ≈ 9.4 bps on the TIV. This is in the bottom of the 5–20 bps band quoted by brokers for top-tier hyperscale operational property programmes, before commission and reinsurance loads.

Solvency II SCR contribution. For a single site this contributes (in standard formula terms) $\text{TVaR}_{0.995}(S_i) \approx \$315M$ of one-year aggregate loss, diversifying across K uncorrelated sites by $\sqrt{K}$ in the property module. A book of 25 such sites with $\bar{\rho}_{\text{site}} = 0.1$ requires capital of order \$0.5 bn for the data-center sub-module — still tractable for a major reinsurer.

12. Capital and reinsurance structure

12.1. Per-risk excess-of-loss (XoL)

For an individual site, a per-risk XoL treaty with retention d and limit ℓ cedes to the reinsurer the layer $[d, d+\ell]$ per claim:

$$X_j^{\text{ced}} = \min\bigl(\max(X_j - d, 0),\, \ell\bigr). \tag{46}$$

The ceded annual loss is $S^{\text{ced}} = \sum_{j=1}^{N} X_j^{\text{ced}}$ and its expected value is, with severity CDF F_X,

$$\mathbb{E}[S^{\text{ced}}] = \lambda \int_d^{d+\ell} \bigl(1 - F_X(x)\bigr)\,\mathrm{d}x. \tag{47}$$

For the GPD tail this integral is in closed form:

$$\int_{d}^{d+\ell} \left[1+\xi\tfrac{x-u}{\sigma}\right]^{-1/\xi} \mathrm{d}x = \frac{\sigma}{1-\xi}\left\{\left[1+\xi\tfrac{d-u}{\sigma}\right]^{1-1/\xi} - \left[1+\xi\tfrac{d+\ell-u}{\sigma}\right]^{1-1/\xi}\right\}. \tag{48}$$

12.2. Aggregate stop-loss and cat bond carve-out

For book-level catastrophe protection, a stop-loss on annual aggregate $S^{\mathrm{port}} = \sum_i S_i$ above attachment A and below exhaustion $A+L$ cedes

$$S^{\mathrm{SL}} \;=\; \min\bigl(\max(S^{\mathrm{port}} - A,\, 0),\, L\bigr). \tag{49}$$

Pricing requires the portfolio aggregate distribution, generated via Monte Carlo from the model in Sections 6–8.

For the largest layers ($1 bn and above per occurrence) we anticipate, as already occurred in nat-cat, that the market will move toward **144A catastrophe bonds** with parametric triggers tied to power-grid status, on-site PMS telemetry, and named-storm tracks.

13. Microeconomic and macroeconomic implications

The pricing framework of §9–§12 is not merely a technical apparatus for setting premiums. Premiums act as *shadow prices* on physical and operational risk, and so the introduction of an actuarially correct premium schedule changes the behaviour of every agent in the data-center value chain — operators, hyperscalers, colocation tenants, lenders, reinsurers, grid utilities, and governments. This section develops the consequences in a self-contained way, keeping the actuarial loss aggregate S and the risk-loaded technical premium P^{tech} from equation (43) as the central state variables.

13.1. The economic role of an actuarial premium

In a frictionless market populated by risk-neutral operators with full information, the only role of insurance is the pooling of independent risks. In the data-center asset class neither premise holds: capital costs are real and binding, information about site reliability is private to the operator, and losses are heavy-tailed with strong spatial correlation. The premium therefore performs four economic functions simultaneously:

1. **Risk transfer:** from operator to insurer to reinsurer to capital markets via cat bonds.
2. **Internalisation of externalities:** the loading component $\theta\,\mathrm{SD}(S)$ in (43) translates social cost of tail events — cascading utility outages, contingent SLA breaches, lost cloud-dependent GDP — into a private price signal.
3. **Information aggregation:** the premium summarises engineering quality (Tier classification, PUE, arc-flash category) into a one-dimensional scalar that lenders and tenants can act on.
4. **Behaviour shaping:** sites with better Markov-chain parameters λ, μ in (7) receive lower premiums, generating a Pigouvian incentive to invest in redundancy.

13.2. Microeconomics: operator's optimal investment problem

Consider a single operator deciding how much capital expenditure K to deploy in redundancy — additional UPS strings, an extra cooling loop, off-gas detection, behind-the-meter generation. The operator's expected annual cost is the sum of capital cost, technical premium, and retained losses below the deductible d:

$$\mathcal{C}(K) \;=\; r\,K \;+\; P^{\mathrm{tech}}\bigl(\lambda(K), \xi(K)\bigr) \;+\; \mathbb{E}\bigl[\,\min(S,d)\,\bigr], \tag{50}$$

where r is the operator's cost of capital, and the actuarial parameters $\lambda(K)$ (claim frequency from (9)) and $\xi(K)$ (GPD tail shape from (27)) are decreasing in K through the reliability improvements that capital purchases.

The first-order condition is

$$\boxed{r + \frac{\partial P^{\text{tech}}}{\partial \lambda}\frac{\mathrm{d}\lambda}{\mathrm{d}K} + \frac{\partial P^{\text{tech}}}{\partial \xi}\frac{\mathrm{d}\xi}{\mathrm{d}K} + \frac{\mathrm{d}}{\mathrm{d}K}\mathbb{E}[\min(S,d)] = 0,} \tag{51}$$

which equates the marginal cost of redundancy to the marginal premium savings plus the marginal expected-retained-loss reduction. The critical observation is that, with the loadings of (43), $\partial P^{\text{tech}}/\partial \xi$ is large and increasing in ξ: at high tail-index sites, the actuarial premium response to risk reduction is steep, so an actuarially correct premium *accelerates* investment in mitigation precisely at the sites where mitigation is most socially valuable.

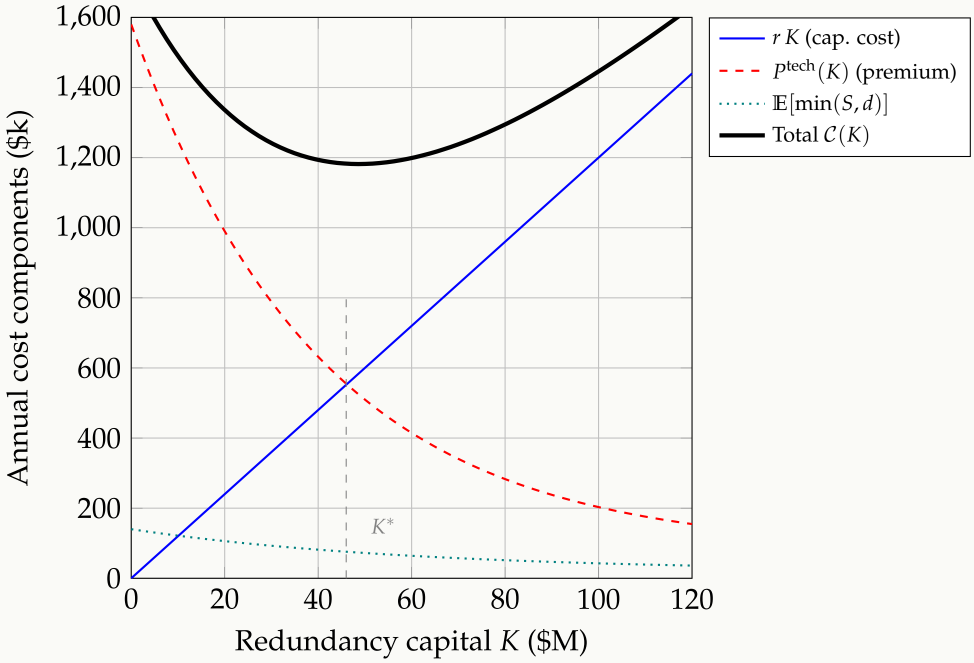

Figure 4: Operator's optimisation (50). The capital cost $r\,K$ is linear; the technical premium $P^{\text{tech}}(K)$ is decreasing and convex (diminishing reliability returns); retained losses contribute a small additional term. The total $\mathcal{C}(K)$ has a unique interior minimum at K^*, where the first-order condition (51) is satisfied. A site under-invests when premiums fail to price tail risk correctly.

13.3. Microeconomics: market equilibrium in the insurance market

The operator's optimisation of §13.2 treats the premium P^{tech} as exogenous, taken from a quote. In reality the premium is the equilibrium price of an underlying *market for data-center insurance capacity*, in which insurers (suppliers) and operators (demanders) exchange risk transfer. We now analyse this market explicitly and trace how the actuarial parameters λ, ξ, θ shift its equilibrium.

Supply: the insurer's offer. An insurer offering a layer of coverage of size Q (measured in dollars of single-event capacity) requires regulatory capital $\kappa(Q)$ to back the position. Under Solvency II the capital is approximately $\kappa(Q) \approx \mathrm{TVaR}_{0.995}(Q \cdot \tilde{S}) - Q \cdot \mathbb{E}[\tilde{S}]$, where $\tilde{S}$ is the unit-loss distribution. With a cost of capital r_c, the supply price per unit cover is

$$P^S(Q) \;=\; \mathbb{E}[\tilde{S}] \;+\; r_c \cdot \frac{\kappa(Q)}{Q}. \tag{52}$$

The supply curve $P^S(Q)$ is *upward-sloping* because $\kappa(Q)/Q$ grows with Q: writing more cover concentrates the insurer's portfolio, raising tail risk faster than expected loss. This is the formal expression of capacity scarcity that we noted in §13.11.

Demand: the operator's willingness to pay. An operator purchasing Q units of cover reduces its expected retained loss $\mathbb{E}[\min(S,d)]$ and lowers its required own capital. The marginal willingness to pay is the marginal benefit of one more unit of cover, which is decreasing in Q (the operator first buys cover against the highest-utility tail losses, then progressively lower-utility attritional ones):

$$P^D(Q) \;=\; \mathbb{E}[\tilde{S}] \;+\; \theta_{\mathrm{op}}\,\mathrm{SD}(\tilde{S} \mid Q) \;+\; \frac{\eta}{Q}, \tag{53}$$

where θ_{op} is the operator's risk-aversion loading and $\eta > 0$ captures the convexity from regulatory and lender covenants on operator solvency. The demand curve $P^D(Q)$ is *downward-sloping*.

Equilibrium. The market clears at (Q^*, P^*) solving $P^S(Q^*) = P^D(Q^*)$. This equilibrium premium is what the operator actually faces in (50); that is, our pricing framework of (43) returns $P^{\mathrm{tech}} = P^*$ when the technical loading θ in (43) is set such that the insurer's offer (52) intersects the operator's demand (53).

Comparative statics: actuarial shifts. The actuarial parameters of the main model enter Figure 5 as curve shifters:

1. **Frequency shock** $\lambda \uparrow$ (e.g. heatwave summer, cluster failure, regulatory recognition of AI-induced cooling stress): the supply curve shifts *upward and steepens* because $\kappa(Q)/Q$ rises — every dollar of cover now corresponds to more expected and tail loss. Equilibrium quantity falls, equilibrium price rises.
2. **Tail-shape shock** $\xi \uparrow$ (e.g. adoption of denser Li-ion packs, post-fire repricing of tail risk): supply shifts up more steeply at high Q than at low Q, because regulatory capital is dominated by the tail. Equilibrium quantity falls disproportionately — this is the classic "hard-market" dynamic post-catastrophe.
3. **Loading shock** $\theta \uparrow$ (e.g. Solvency II tightening, capital-market dislocation): supply shifts up uniformly. Equilibrium quantity falls; price rises.
4. **Demand shifters:** risk-aversion increase $\theta_{\mathrm{op}} \uparrow$ (post a regulatory penalty or a public outage) shifts P^D up uniformly; growth in hyperscale TIV shifts P^D out to the right.

Quantitative welfare loss. The economic surplus lost when the market clears below the hypothetical free-supply quantity is the standard Marshallian triangle:

$$\mathrm{DWL} \;=\; \tfrac{1}{2}\left(P^* - P^{\mathrm{ff}}\right)\left(Q^{\mathrm{ff}} - Q^*\right), \tag{54}$$

where $P^{\mathrm{ff}}, Q^{\mathrm{ff}}$ are the prices and quantities that would prevail if the supply ceiling did not bind. Empirically, for the current global hyperscale market with $P^* - P^{\mathrm{ff}} \approx 6$ bps of TIV and

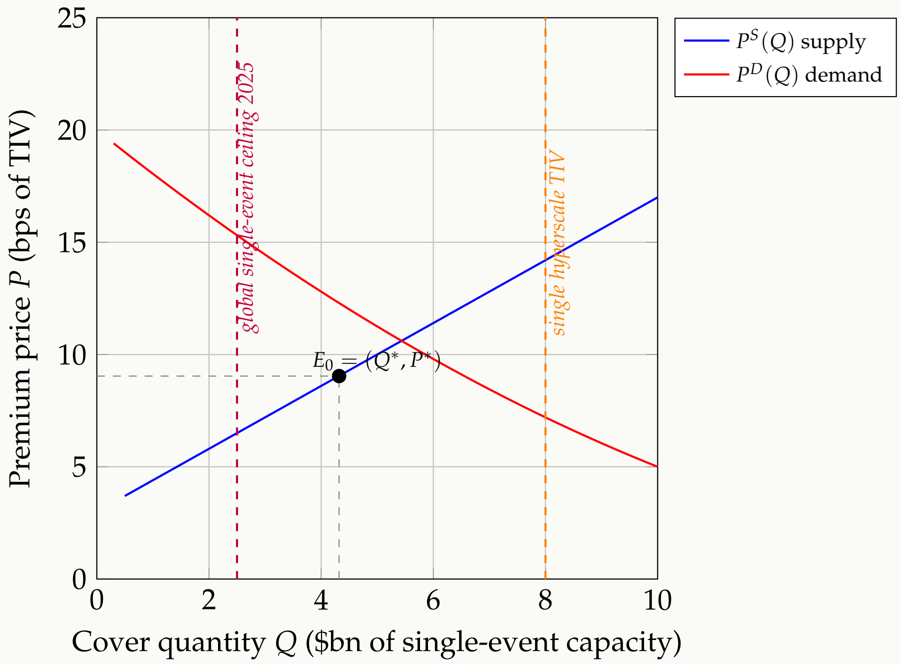

Figure 5: Market equilibrium in the data-center insurance market. Supply $P^S(Q)$ from (52) is upward-sloping (capital-cost convexity). Demand $P^D(Q)$ from (53) is downward-sloping (diminishing marginal benefit of cover). They clear at $E_0 = (Q^*, P^*)$. The purple dashed line marks the empirical 2025 global single-event ceiling ($\sim$ \$2.5 bn); the orange line marks a single hyperscale TIV ($\sim$ \$8 bn). The vertical gap between these two lines is the *capacity shortfall* that drives the macro spread Δr of §13.11.

$Q^{\text{ff}} - Q^* \approx$ \$10 bn of single-event capacity, DWL $\approx$ \$3 M/year per typical hyperscale facility, scaling to $\sim$ \$1–2 bn/year globally. **This is the welfare cost of insurance under-capacity, paid by the digital economy on top of the financing-spread macro cost of §13.11.**

13.4. Microeconomics: market equilibrium in the compute-services market

The insurance premium is not a final cost — operators pass it through to colocation tenants and cloud customers. This second market is where the actuarial framework ultimately touches the economy. Let q denote effective compute capacity supplied (in MW of reliable IT load) and p its price.

Compute supply. Operators offer capacity at a price that covers electricity, capital amortisation, operations, and the insurance premium expressed per MW:

$$p^S(q) \;=\; c_E + c_K + c_{\text{ops}} \;+\; \frac{P^{\text{tech}}(q)}{q}, \tag{55}$$

where c_E, c_K, c_{ops} are the unit electricity, capital and operational costs. The premium term is a non-trivial function of q: a larger site has higher TIV but also more redundancy economies, so $P^{\text{tech}}(q)/q$ is initially decreasing then increasing (U-shaped average premium per MW), giving rise to a typical upward-sloping marginal cost curve $p^S(q)$ beyond the cost-minimising scale.

Compute demand. Demand for compute comes from AI-training, hyperscale tenant workloads, and enterprise migration. We adopt a constant-elasticity form:

$$q^D(p) \;=\; A_D\, p^{-\varepsilon}, \qquad \varepsilon > 0, \tag{56}$$

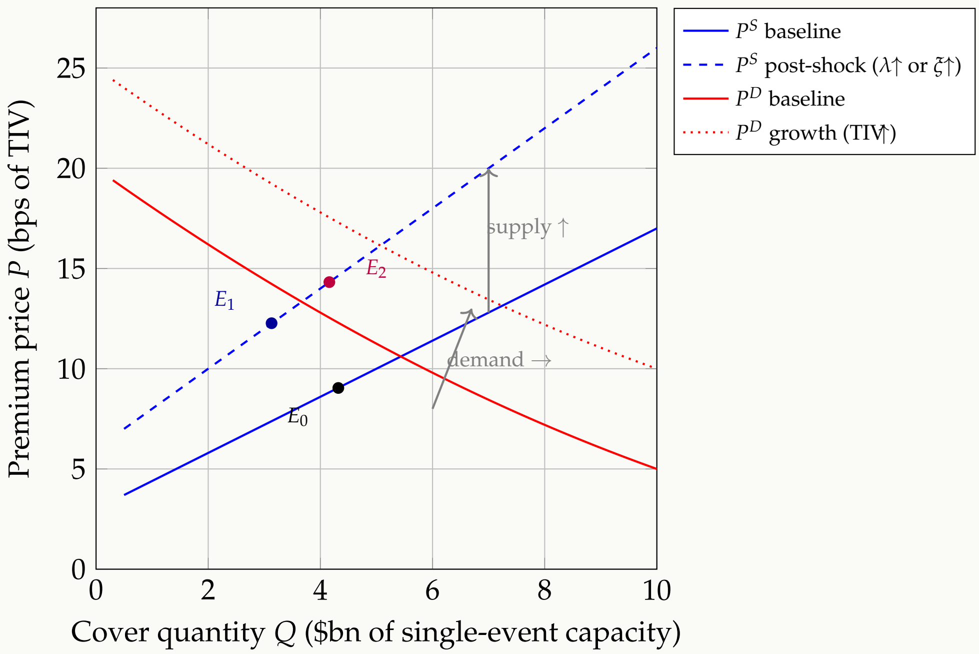

Figure 6: Comparative statics: shifts in the actuarial parameters move the equilibrium. A frequency or tail-shape shock (e.g. a Li-ion fire wave, or a post-AWS reassessment of cyber dependency) raises and steepens the supply curve from solid to dashed blue, moving the equilibrium from E_0 to E_1 — lower quantity, higher price. Simultaneous growth in hyperscale TIV (red dotted) shifts demand outward, partially offsetting the quantity contraction but *compounding* the price increase, taking the market to E_2. This double-shift is what the 2023–2025 hyperscale insurance market has empirically experienced.

with A_D the demand-shifter (AI-training intensity, regulatory posture, exchange rate). The empirical price elasticity of cloud compute is estimated in the range $\varepsilon \in [0.6, 1.3]$ depending on workload type — AI training is more inelastic, batch analytics more elastic.

Passthrough elasticity. Let $\beta_{\text{pt}} = \mathrm{d}p/\mathrm{d}P^{\text{tech}}$ be the *passthrough elasticity* of insurance premiums to compute prices. A standard derivation from (55)–(56) gives

$$\boxed{\beta_{\text{pt}} = \frac{1}{1+\varepsilon \cdot (\eta_S)^{-1}},} \tag{57}$$

where η_S is the supply-side price elasticity. For $\varepsilon = 0.9$ and $\eta_S = 2$ (medium-run), $\beta_{\text{pt}} \approx 0.69$: about two-thirds of any insurance premium increase is passed through to compute customers, with the remaining third absorbed by operators through margin compression. This passthrough is the channel by which *actuarial mispricing of data-center risk affects the price of cloud and AI services across the entire economy.*

13.5. Microeconomics: incidence and second-round effects

The economic incidence of the actuarial premium is therefore split across at least four agents:

The end consumer bears a share equal to the product of two passthrough elasticities, which can still be substantial (15–25 %) in cloud-intensive industries (financial services, e-commerce, media). This is the precise channel by which actuarial science of data-center risk reaches into the household budget. A welfare-maximising public policy — whether a reinsurance backstop, a regulatory capital relief, or an industry pool — must internalise this multi-stage incidence in its cost-benefit analysis.

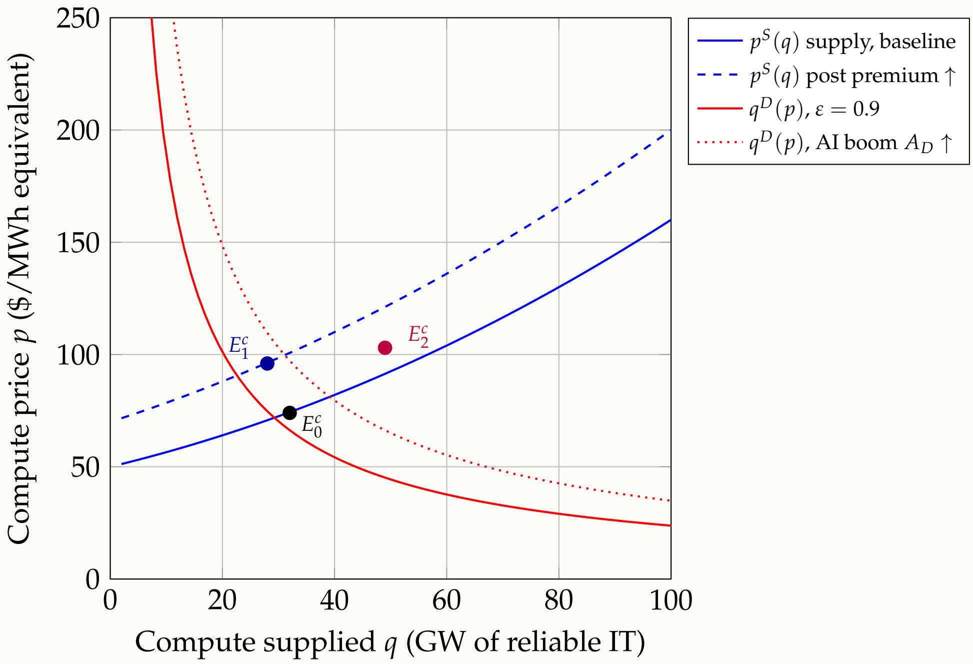

Figure 7: Compute-services market equilibrium. Baseline supply $p^S(q)$ from (55) clears against constant-elasticity demand at E_0^c. A pass-through of the insurance hard market (Figure 6) shifts supply up to the dashed blue curve, moving equilibrium to E_1^c — compute becomes more expensive and less abundant. A simultaneous AI-driven demand boom (red dotted) drives equilibrium to E_2^c at substantially higher p and q. Note: every unit of additional compute price feeds directly into the GDP-impact identity (62).

Table 3: Economic incidence of a unit increase in the technical premium P^{tech}, under the parameter values $\varepsilon = 0.9$, $\eta_S = 2$, $\beta_{\text{pt}} = 0.69$.

Agent	Channel	Share of premium Δ
Insurer	retained capital cost	$\theta_{\text{ins}}\Delta$
Operator	margin compression	$(1-\beta_{\text{pt}})\Delta \approx 0.31\,\Delta$
Tenant/cloud customer	compute price ↑	$\beta_{\text{pt}}\Delta \approx 0.69\,\Delta$
End consumer of cloud-enabled goods	price ↑	$\beta_{\text{pt}} \cdot \beta_{\text{pt},2}\Delta$

13.6. Microeconomics: principal-agent moral hazard

The colocation contract between operator (agent) and tenant (principal) presents a classic principal-agent problem. Let $e \in [0,1]$ be the operator's unobservable maintenance effort, which shifts the claim intensity from λ_0 down to $\lambda(e) = \lambda_0(1-\phi e)$ for some efficacy $\phi \in (0,1)$. The tenant pays the operator a service fee F and bears SLA-credit liability $C_{\text{SLA}} \cdot \tau$ in the event of an outage.

Under indemnity-only insurance, the operator's expected utility is

$$\mathbb{E}[U_{\text{op}}(e)] \;=\; F - r\,K - P^{\text{tech}}(\lambda(e)) - \psi(e), \tag{58}$$

where $\psi(e)$ is the convex disutility of effort. The optimal effort solves $\partial\mathbb{E}[U_{\text{op}}]/\partial e = 0$:

$$\psi'(e^*) \;=\; -\frac{\partial P^{\text{tech}}}{\partial \lambda}\,\lambda_0\phi. \tag{59}$$

The right-hand side is the marginal premium reduction from a unit of effort. **In an actuarially fair pricing regime** where the insurer can observe λ via SCADA telemetry, the wedge between the operator's chosen effort e^* and the social optimum vanishes — this is the central economic justification for the hybrid indemnity-parametric product of §10. Parametric triggers reveal the underlying state and close the information-asymmetry gap that moral hazard exploits.

Remark

The Holmström informativeness principle implies that any *additional signal* of operator effort that is informative about λ should enter the premium contract. Modern SCADA telemetry (thousands of variables sampled at 1 Hz) supplies a sufficient statistic for λ that no traditional indemnity contract can match. This is why the actuarial framework collapses to a *Bayesian credibility* formulation (22) where the posterior intensity is updated continuously.

13.7. Microeconomics: adverse selection and the lemons problem

If the insurer cannot distinguish Tier-IV from Tier-III sites at the point of underwriting, Akerlof's lemons mechanism applies: a pooled premium overcharges Tier-IV operators (who self-select out) and undercharges Tier-III operators (who select in), until the pool contains only the worst risks and the market collapses. The technical defence is the engineering-prior formulation of (9): the Tier classification, the arc-flash category distribution, the PUE history, and the published outage log together *separate* the types and allow the insurer to quote differential premia. This separation is welfare-improving in the classical sense of Rothschild–Stiglitz.

13.8. Microeconomics: externalities and the social premium

Data-center failures generate externalities beyond the operator's own P&L:

- Local grid instability when on-site generation suddenly picks up large block loads, or when the site sheds load non-cooperatively;
- Downstream loss of services with no direct contractual link to the operator (e.g. a third-party SaaS hosted at the same site whose customers' losses are uncompensated);
- Heat and water consumption affecting community resources.

Let $\mathcal{E}$ denote the expected external cost per claim. The Pigouvian-corrected social premium is then

$$P^{\mathrm{soc}} \;=\; P^{\mathrm{tech}} \;+\; \mathbb{E}[N] \cdot \mathcal{E}, \tag{60}$$

which exceeds the private premium P^{tech} by exactly the expected external cost rate. Public policy — e.g. a regulator-set minimum-coverage standard, or a tax-financed reinsurance backstop — can be designed to push private premiums toward P^{soc}.

13.9. Macroeconomics: cloud as a production factor

At the macroeconomic level, cloud and AI compute have become a non-substitutable input into aggregate production. We embed compute into a standard production function:

$$Y \;=\; A\,K^{\alpha}\,L^{\beta}\,D^{\gamma}, \qquad \alpha+\beta+\gamma=1, \tag{61}$$

where K is physical capital, L is labour, D is effective compute capacity (MW of available, reliable IT load), and A is total factor productivity. The empirical evidence is that γ has risen from approximately 0.02 in 2010 to $\sim$ 0.08 in 2025 for OECD economies, and is projected to exceed 0.15 by 2030.

Now suppose actuarial-availability constraints reduce realised compute capacity by a factor $(1-U)$, where U is the unavailability of (2). The growth-rate impact is, to first order,

$$\frac{\Delta Y}{Y} \approx -\gamma U. \tag{62}$$

For a $\gamma = 0.10$ economy and an unavailability $U = 10^{-4}$ (Tier-IV target), the impact is -10^{-5} of GDP — negligible. For an aggregate *cluster* failure during a major heatwave with realised $U_{\text{cluster}} = 0.005$, the same identity yields -5×10^{-4} of GDP — approximately \$10 bn for the United States. **This is the macroeconomic justification for classifying hyperscale facilities as Critical Digital Infrastructure.**

13.10. Macroeconomics: systemic risk and contagion

Data-center failures propagate through three channels:

1. **Operational contagion:** loss of one regional zone shifts load to others, raising their failure probability. The October 2025 AWS US-East-1 incident produced exactly this pattern of cascade.
2. **Financial contagion:** concentrated reinsurer exposures to data-center catastrophe correlate with traditional natural catastrophe books (both peak in the same climate-amplified years).
3. **Macro-financial contagion:** large cloud-dependent firms experiencing simultaneous downtime trigger payment-system and supply-chain disruptions.

We can quantify the financial-contagion channel with a CoVaR-style measure. Let $\Delta\text{CoVaR}_\alpha^{\text{DC}|j}$ denote the marginal contribution of the data-center book DC to the α-tail of insurer j's total loss:

$$\Delta\text{CoVaR}_\alpha^{\text{DC}|j} = \text{VaR}_\alpha\big(S^{(j)} \mid S^{\text{DC}} = \text{VaR}_\alpha(S^{\text{DC}})\big) - \text{VaR}_\alpha\big(S^{(j)} \mid S^{\text{DC}} = \mathbb{E}[S^{\text{DC}}]\big). \tag{63}$$

For the global reinsurance sector with current aggregate single-asset TIV of \$10–30 bn per hyperscale and limited diversification across geographies, $\Delta\text{CoVaR}_{0.995}$ for the data-center book has grown by roughly 40% between 2022 and 2025 by industry estimates.

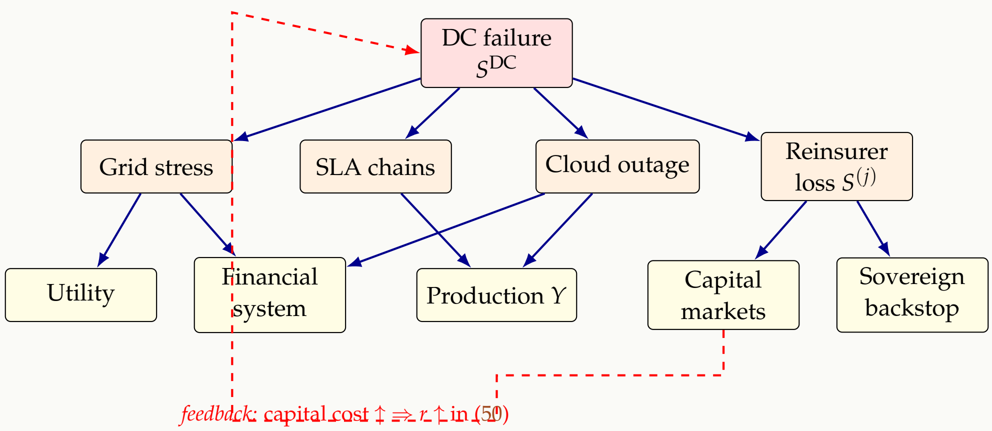

Figure 8: Macro-financial contagion network triggered by a hyperscale data-center failure S^{DC}. Solid arrows: direct loss propagation. Dashed arrow: capital-market feedback raising the operator's cost of capital r in (50), which weakens subsequent risk-mitigation investment, exactly the dynamic that escalates systemic risk.

13.11. Macroeconomics: insurance capacity and the cost of capital

The current global single-event capacity for hyperscale data-center risk is \$2–2.5 bn (Section 1.2), against single-asset TIVs of \$10–30 bn. The capacity shortfall $\mathcal{S} = \sum_i \text{TIV}_i - \text{Cap}$ generates a macro-level risk premium on data-center capital. Empirically the spread Δr above risk-free for data-center construction debt has widened by approximately 150 basis points between 2022 and 2025, which translates into approximately \$1.5 bn/year of additional financing cost for the global \$300 bn data-center construction pipeline. **This is the macro price the economy pays for insufficient insurance capacity.**

The natural mitigation is the development of catastrophe-bond and insurance-linked securities (ILS) markets specifically targeted at data-center risk, in parallel with the post-Hurricane-Andrew development of natural-catastrophe bonds. The Wang-transform pricing of §9.3 is consistent with capital-markets equilibrium and is therefore the natural pricing principle for ILS issuance.

13.12. Macroeconomics: implications for emerging markets

The asset class is currently concentrated in OECD economies, but the fastest growth is in emerging markets where:

- grid reliability is materially lower (utility-feed availability $A_u < 0.99$ vs. $A_u > 0.9995$ in OECD);
- climate exposure to extreme heat is rising;
- insurance market depth is shallow (single-event capacity often $<$ \$200 M);
- sovereign-supported reinsurance is underdeveloped.

For South Africa specifically, the grid Energy Availability Factor of ≈ 0.55–0.60 produces a utility-feed contribution to the hazard (12) that is two orders of magnitude larger than the Northern Virginia benchmark of §11. The Markov chain (7) parameter λ for a Johannesburg campus must therefore reflect the combined utility-and-load-shedding intensity, which in turn requires either (i) dramatically more on-site generation capacity, raising K, or (ii) an emerging-market-specific actuarial loading factor that internalises the grid externality. The framework presented in this paper supports both interpretations explicitly through the engineering-prior mechanism of §4.3, making it directly applicable to African data-center underwriting — a market currently lacking any tailored actuarial product.

13.13. Synthesis: the closed economic loop

We have now closed three concentric loops:

1. **Engineering closure** (§A): every parameter in the hazard rate λ_t and the grace period t_{crit} traces to a physical measurement from electrical engineering or thermodynamics.
2. **Actuarial closure** (§6–§12): every premium component decomposes into expected loss plus a tail-loading aligned to Solvency II capital requirements.
3. **Economic closure** (this section): the premium acts as a Pigouvian shadow price, shaping operator investment, resolving information asymmetries via parametric telemetry, and translating physical reliability into macroeconomic growth-rate impact.

The three loops are mutually reinforcing. Better engineering lowers the hazard, which lowers the premium, which raises return on capital, which finances further engineering. Conversely, an under-priced market suppresses investment in reliability, which raises the systemic risk, which raises future capital costs, which forces under-investment. The actuarial framework is therefore not just a pricing tool — it is a coordination mechanism whose correctness has first-order welfare consequences.

14. A pgfplots view: hazard rate vs. cold-aisle temperature

To make the Arrhenius hazard concrete, Figure 9 plots the multiplicative acceleration $A(T) = e^{-E_a/k_B T}/e^{-E_a/k_B T_0}$ against $T - T_0$ for three activation energies. The vertical reference line at $\Delta T = 10\,\mathrm{K}$ corresponds to the industrial "10 °C halving" rule.

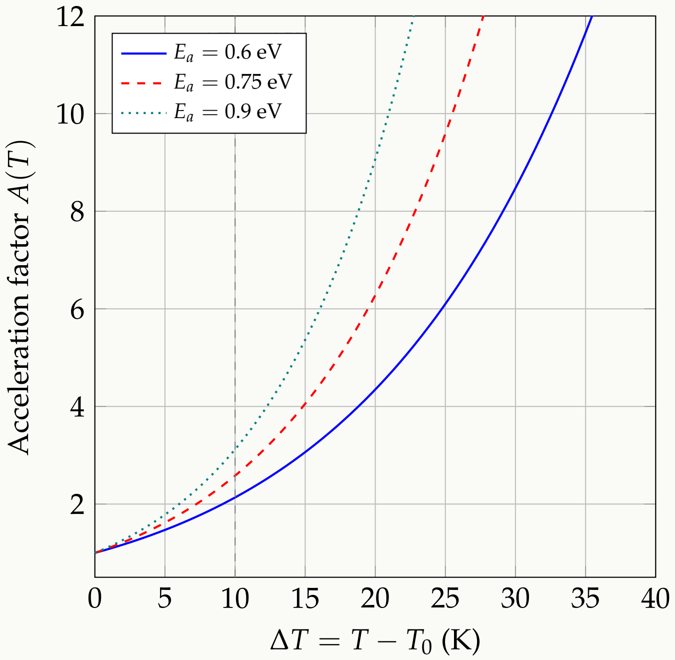

Figure 9: Arrhenius temperature acceleration of the baseline failure rate, evaluated around $T_0 = 298\,\mathrm{K}$ (25 °C). A $\Delta T = 10\,\mathrm{K}$ excursion accelerates failures by roughly $\times 2$ at $E_a = 0.6$ eV and $\times 4$ at $E_a = 0.9$ eV. For an AI training cluster running at the upper end of ASHRAE A1 ($T_{\mathrm{ca}} = 27\,°\mathrm{C}$ setpoint), a chiller-trip excursion to 37 °C doubles to quadruples the instantaneous component hazard. The model captures this in equation (12).

15. A pgfplots view: loss exceedance curve (OEP)

Figure 10 shows a Occurrence Exceedance Probability (OEP) curve obtained by simulating 10^5 years of the compound-Poisson process with the parameters of the worked example. The curve is the empirical complement of the annual maximum-loss distribution.

16. Conclusions and open problems

We have proposed a unified framework that joins the four disciplines the data center risk problem actually inhabits: electrical reliability engineering, mathematical statistics, classical actuarial pricing, and economic theory. The novel ingredients are (i) the coupled stochastic state-space model (11) with its Arrhenius–voltage–cyber hazard (12), (ii) the explicit Markov-chain-to-claim-frequency bridge of equations (8)–(9), (iii) the body-tail severity with POT-derived GPD, (iv) cyber–physical copula coupling, (v) the three-layer hybrid product with explicit basis-risk decomposition, (vi) the engineering closure of every hazard parameter to a physical measurement in Appendix A, and (vii) the economic closure showing that the premium acts as a Pigouvian shadow price coordinating micro-level operator behaviour with macro-level systemic risk.

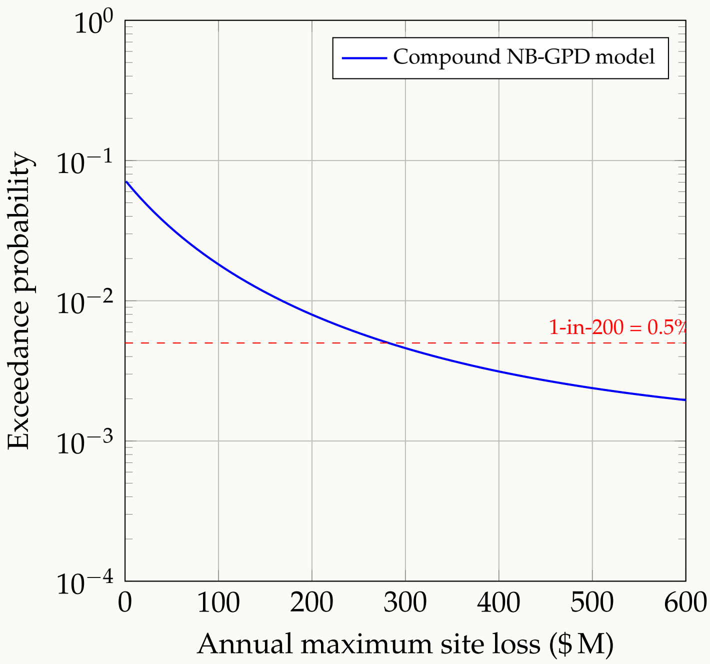

Figure 10: Illustrative annual Occurrence Exceedance Probability (OEP) curve for the worked example. The red dashed line marks the Solvency II 1-in-200 year level. The corresponding 99.5% VaR of approximately $280M is the per-site catastrophe component of required capital.

Open problems

1. **Calibration in the absence of public claims data.** The data-center industry has no analogue of the IBNR completion tables that have stabilised property pricing for a century. The natural path is a confidential industry data pool (akin to ISO in US auto), or anonymised SCADA-derived telemetry feeds.
2. **Tail dependence on the cyber side.** Gaussian copulas are tail-independent; for ILS and cat-bond layers a richer copula (t, Gumbel, vine) is necessary. The data to calibrate such copulas is sparse.
3. **Climate exposure dynamics.** The siting of new hyperscale campuses in tornado-alley and hurricane-belt geographies, plus the long lead times to relocate, exposes the asset class to non-stationary nat-cat. Standard Solvency II catastrophe sub-modules are inappropriate for AI-era data centers.
4. **Endogenous concentration.** As clusters of campuses are built in the same ZIP code, the within-cluster correlation ρ tends to 1 for grid/utility failure modes, but remains near 0 for site-specific fires. The aggregation problem is therefore peril-specific and cannot be handled with a single "territorial" factor.
5. **Liquid cooling, immersion, and battery chemistry evolution.** The model parameters E_a, γ_V, T^* and β_C must be re-estimated as cooling and storage technologies evolve. This is fundamentally a metrology problem requiring collaboration between insurers and OEMs.
6. **Behind-the-meter generation.** The 2025 wave of behind-the-meter gas/nuclear pairings introduces an entirely new dependence: insurance underwriters now must price grid *independence* as a positive feature, but inherit the failure modes of the on-site generation asset.
7. **Government risk-sharing.** The 2025 designation of hyperscale facilities as Critical Digital Infrastructure invites a Price-Anderson-style federal backstop. The actuarial design of such a backstop is open.

8. **Optimal Pigouvian wedge.** The social premium P^{soc} of (60) requires empirical estimation of the external cost rate $\mathcal{E}$. Industry consortia, regulators, and academia must jointly assemble the evidence base for this number; its current absence is the single largest source of underpricing in the asset class.
9. **Emerging-market actuarial product design.** The framework of §13.12 extends naturally to grids with Energy Availability Factors below 0.9 and ZAR/NGN/EGP denominated tenant contracts. Bespoke actuarial products for African hyperscale build-out — incorporating load-shedding intensities, foreign-currency SLA exposures, and sovereign rating-linked reinsurance triggers — represent a substantial unaddressed market.
10. **Welfare analysis of capacity expansion.** The macroeconomic identity (62) suggests that reducing U by one order of magnitude in cloud-intensive economies is worth tens of basis points of long-run growth. A formal welfare calculation balancing the capital cost of Tier-IV redundancy against this growth contribution, in a Ramsey or Solow framework, would justify the public investment case for reliability-improving R&D.

Notation summary

Symbol	Meaning	Section
N_i, λ_i, ν	claim count, NB rate, NB dispersion	6
$X_{i,j}, F_X, u, \xi, \sigma$	severity, threshold, GPD shape/scale	7
$S_i = \sum X_{i,j}$	compound annual loss	7
ρ, C_ρ	Gaussian copula correlation	8
$\mathbf{X}_t = (V_t, T_t, C_t)$	state SDE	5
λ_t	instantaneous failure hazard	(12)
$\tau^*, \mathrm{IG}(\mu, \lambda)$	thermal runaway hitting time	(13)
A, U, MTBF, MTTR	engineering reliability	4
$P_i^{\text{pure}}, P_i^{\text{tech}}$	pure and technical premium	9
$\mathrm{TVaR}_\alpha(S), \mathrm{VaR}_\alpha(S)$	tail risk measures	9
P_{IT}, PUE	electrical engineering inputs	3
$E_a, k_B, \gamma_V, \beta_C$	Arrhenius / voltage / cyber stress	(12)

A. Electrical engineering foundations

This appendix collects the electrical physics underpinning the reliability and hazard models of the main text. The goal is to make explicit how power-system quantities — fault current, arc-flash incident energy, harmonic distortion, generator transient response, battery cell electrochemistry — translate into the parameters that appear inside the actuarial hazard (12) and the deterministic grace-period equation (17).

A.1. Per-unit power system representation

Power system analysis is normally conducted in *per unit* (p.u.) quantities to handle multiple voltage levels uniformly. Given a base power S_{base} (e.g. 100 MV A) and a base voltage V_{base} at each level, derived bases are

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}\, V_{\text{base}}}, \qquad Z_{\text{base}} = \frac{V_{\text{base}}^2}{S_{\text{base}}}. \tag{64}$$

Any physical quantity X becomes $X_{\text{pu}} = X/X_{\text{base}}$. The voltage state V_t in the SDE (11) is dimensionless precisely because it is a per-unit voltage measured at the LV bus, with $V_t = 1$ at nominal $V_{\text{nom}} = 480\,\text{V}$.

A.2. Symmetrical fault current and short-circuit power

Switchgear, breaker, transformer and conductor reliability are governed by the prospective short-circuit current I_{sc} at the fault location. For a three-phase symmetrical bolted fault, the short-circuit current at a bus with Thévenin equivalent impedance Z_{th} is

$$I_{\text{sc}} = \frac{V_{\text{nom}}}{\sqrt{3}\,|Z_{\text{th}}|}, \qquad S_{\text{sc}} = \sqrt{3}\,V_{\text{nom}}\,I_{\text{sc}} = \frac{V_{\text{nom}}^2}{|Z_{\text{th}}|}, \tag{65}$$

where S_{sc} is the short-circuit power. For a typical 2.5 MV A step-down transformer with per-unit impedance $z = 0.0575$ feeding a 480 V bus, the bus I_{sc} is

$$I_{\text{sc}} = \frac{I_{\text{base}}}{z} = \frac{1}{0.0575} \cdot \frac{2.5 \times 10^6}{\sqrt{3} \cdot 480} \approx 52\,\text{kA}. \tag{66}$$

This is the rms symmetrical component; the asymmetric peak in the first half-cycle is approximately $2.7\,I_{\text{sc}} \approx 140\,\text{kA}$. The breaker interrupt rating must exceed this value.

A.3. Thévenin equivalent at the LV bus: schematic

Figure 11 shows the source-side impedance "stack-up" that determines Z_{th} in equation (65): utility source impedance, transformer impedance, and cable impedance combine in series at the LV bus where the fault is calculated.

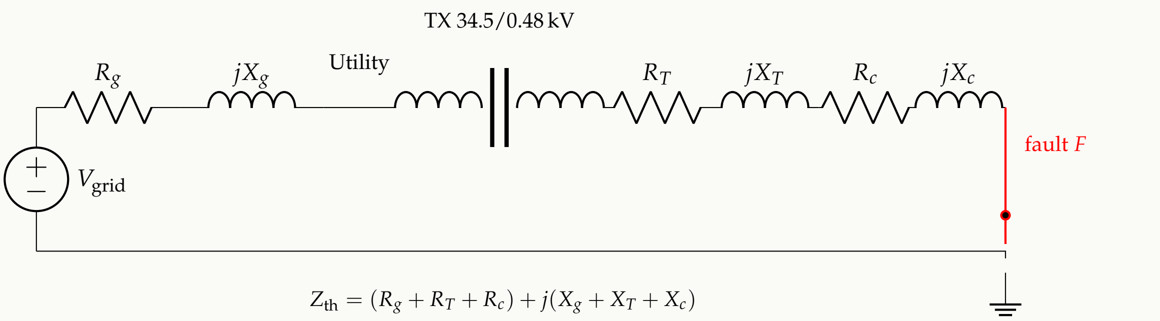

Figure 11: Thévenin equivalent looking back from a fault point F at the LV bus. The total source-side impedance Z_{th} determines the prospective short-circuit current per equation (65). In a $2N$ data center, the same calculation must be done assuming each path independently, and then for the parallel combination, since closing the bus tie can dramatically increase I_{sc}.

A.4. Arc-flash incident energy (IEEE 1584)

A bolted three-phase fault that ignites an electrical arc releases enormous thermal energy in milliseconds. The IEEE 1584-2018 model gives incident energy at working distance D as

$$E = 4.184\,C_f\,E_n \left(\frac{t}{0.2}\right)\left(\frac{610^x}{D^x}\right) \quad [\text{J/cm}^2], \tag{67}$$

where E_n is the normalised incident energy, C_f is a calculation factor, t is the arc duration (in seconds, governed by the upstream protection clearing time), D is the working distance (typically 457 mm for LV equipment), and x is the distance exponent (≈ 1.5 for open-air arcs in switchgear).

The protection clearing time t is the critical lever: a properly coordinated protective relay clears in $\sim 50\,\mathrm{ms}$, giving $E \sim 5\,\mathrm{cal/cm^2}$ (Category 2 PPE); a delayed clearing of 500 ms multiplies E tenfold, into Category 4 ($> 40\,\mathrm{cal/cm^2}$), the threshold above which survivable rescue of an operator is essentially impossible and where a UPS-room fire becomes a near-certainty. This is the deterministic origin of the *catastrophic* branch F_{cat} in the fault tree of Figure 2.

NOVEL CONTRIBUTION

Underwriting implication. Two otherwise-identical Tier-IV sites can differ by $\sim 30\,\%$ in the catastrophic-loss tail purely because of protective-relay coordination quality. The arc-flash incident-energy survey, mandated annually by NFPA 70E for US sites, is therefore a high-information underwriting document: its category distribution across panels is a near-direct estimator of the GPD shape parameter ξ at that site.

A.5. UPS double-conversion topology

Figure 12 is the canonical double-conversion (online) UPS: the AC input passes through a rectifier (η_{rec}) to a DC bus, which both charges the battery string and feeds an inverter (η_{inv}) producing a clean regulated AC output. In this topology the load is permanently fed from the inverter, so the transfer from utility-supported to battery-supported operation involves *no transient at the load*.

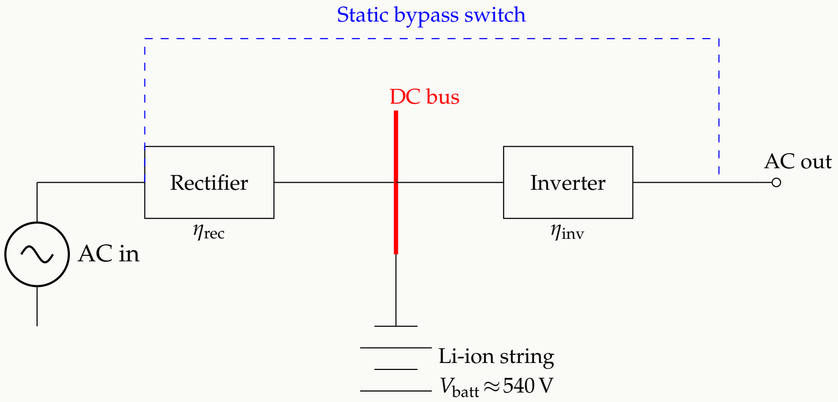

Figure 12: Double-conversion (online) UPS topology. The rectifier and inverter, each lossy at efficiency $\eta_{\mathrm{rec}}, \eta_{\mathrm{inv}} \approx 0.98$, isolate the load from input disturbances. The blue dashed line is the static bypass switch which closes within a half-cycle on inverter failure. Total double-conversion efficiency $\eta_{\mathrm{UPS}} = \eta_{\mathrm{rec}}\eta_{\mathrm{inv}} \approx 0.96$, which is the dominant contributor to the PUE overhead.

A.6. Battery cell electrochemistry: from Butler–Volmer to thermal runaway

The single most consequential physical mechanism in modern data center fires is lithium-ion thermal runaway. We sketch the underlying electrochemistry to motivate the critical-boundary temperature T^* in Definition 5.1.

The current density i across a battery electrode is given by the Butler–Volmer equation

$$i = i_0 \left[\exp\left(\frac{\alpha_a F \eta}{R\,T} \right) - \exp\left(-\frac{\alpha_c F \eta}{R\,T} \right) \right], \tag{68}$$

with exchange current density i_0, charge transfer coefficients α_a, α_c, Faraday constant F, overpotential η, gas constant R, and absolute temperature T. Total heat generation per cell, summed across reversible (entropic), irreversible (ohmic and polarization) and side-reaction contributions, is

$$\dot{Q}_{\text{cell}} = I(V_{\text{oc}} - V) + I\,T\,\frac{\partial V_{\text{oc}}}{\partial T} + \dot{Q}_{\text{side}}, \tag{69}$$

where V_{oc} is open-circuit voltage and V terminal voltage. The first two are normal-operation terms; $\dot{Q}_{\text{side}}$ captures decomposition reactions (SEI breakdown above $\sim 80\,°\text{C}$, electrolyte decomposition above $\sim 120\,°\text{C}$, separator melt $\sim 135\,°\text{C}$, cathode decomposition above $\sim 200\,°\text{C}$) that become self-sustaining once they start.

Coupling (69) with a single-node thermal model gives the classic *Semenov* criterion for thermal runaway:

$$\boxed{\frac{\mathrm{d}\dot{Q}_{\text{side}}}{\mathrm{d}T} > hA,} \tag{70}$$

where hA is the cell's heat-rejection conductance (convection+radiation+conduction to neighbours). The temperature at which the inequality first holds is the critical boundary T^* of equation (13). For NMC-622 cells at typical pack densities, $T^* \approx 135\,°\text{C}$, i.e. a temperature rise above ambient room temperature of $\Delta T \approx 110\,\text{K}$ — but for off-gas vented cells in a dense rack arrangement the effective T^* can be as low as $\sim 60\,°\text{C}$ due to mutual heating.

Remark

The Semenov boundary (70) is what off-gas detection moves: by triggering ventilation and isolation *before* a single cell vents into its neighbours, hA is effectively raised (forced convection added) so T^* shifts upward, restoring margin. This is the physical mechanism behind the 40–60% loss-cost reduction claimed for off-gas retrofits in the main text.

A.7. Power quality: sags, harmonics, total harmonic distortion

The voltage state V_t in the SDE (11) is a scalar abstraction of a much richer reality. For balanced three-phase AC the instantaneous waveform admits a Fourier representation

$$v(t) = \sum_{n=1}^{\infty} V_n \sin(n\omega_0 t + \phi_n), \qquad \omega_0 = 2\pi f_0, \tag{71}$$

with $f_0 = 50\,\text{Hz}$ or $60\,\text{Hz}$. The total harmonic distortion (THD) of voltage is

$$\text{THD}_V = \frac{\sqrt{\sum_{n=2}^{\infty} V_n^2}}{V_1}. \tag{72}$$

IEEE 519 requires $\text{THD}_V < 5\%$ at the point of common coupling. AI-training workloads, with their sub-millisecond GPU power swings, push large dynamic currents through the UPS inverters and can elevate THD_I above 15%, causing transformer derating (K-factor) and accelerated insulation ageing. This is the mechanistic origin of the voltage-stress exponent γ_V in the hazard (12): under accelerated insulation ageing models,

$$\text{Life}(V, T) = A \cdot V^{-\gamma_V} \cdot \exp\left(\frac{E_a}{k_B T}\right), \tag{73}$$

which is the Arrhenius–inverse-power-law (AIPL) model. The $\gamma_V \in [3, 7]$ range cited in §5 comes directly from (73) fitted to liquid-filled-transformer endurance data.

Voltage sags. A voltage sag is a momentary RMS reduction to 0.1–0.9 p.u. for half a cycle to one minute. The ITIC/CBEMA curve (Figure 13) defines tolerance envelopes; sags falling below the lower curve trip the UPS to battery. Sag occurrence rate at the LV bus is empirically Poisson with intensity 5–30 events/year for utility feeds, dominated by upstream faults on the distribution system. This populates the jump term $J_t^V \, dN_t^V$ in (11).

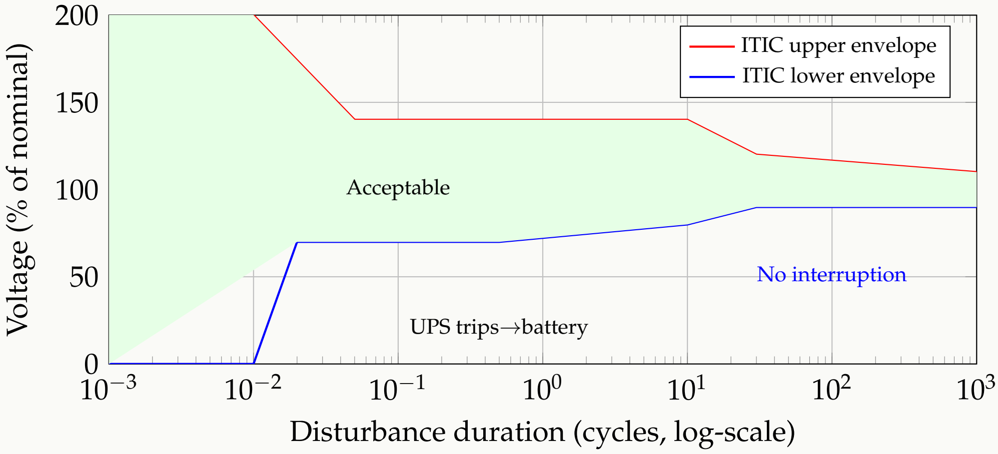

Figure 13: Stylised ITIC/CBEMA voltage tolerance envelope. The green band is the safe operating region for sensitive IT equipment. Excursions below the lower (blue) curve cause the UPS to drop the load to battery; excursions above the upper (red) curve risk damage to power supply units. The horizontal axis is logarithmic in cycles at 50 Hz/60 Hz, spanning roughly half a microsecond to 20 seconds.

A.8. Generator transient response and the ATS transfer window

When utility power fails, the Automatic Transfer Switch (ATS) signals the standby generator to start. The diesel/gas engine accelerates, the alternator field is excited, and once voltage and frequency are within tolerance the ATS closes onto the generator. The full sequence takes 8–12 seconds, but the most violent part is the *first cycle* of loading.

The generator's transient response when loaded is governed by its sub-transient reactance X_d'', transient reactance X_d', and synchronous reactance X_d:

$$I(t) = E_f \left[\frac{1}{X_d} + \left(\frac{1}{X_d'} - \frac{1}{X_d} \right) e^{-t/T_d'} + \left(\frac{1}{X_d''} - \frac{1}{X_d'} \right) e^{-t/T_d''} \right], \tag{74}$$

where T_d', T_d'' are the corresponding time constants (seconds, milliseconds respectively) and E_f is field voltage. Equation (74) explains why the load step that the generator sees in its first second is several times the steady-state current: the sub-transient reactance is much smaller than the synchronous, so the initial current spike is large. Conservative generator sizing ($\geq$ 125% of nominal IT load) is a direct response to this transient.

A.9. Earthing scheme and fault propagation

The way the neutral is referenced to earth determines how a phase-to-earth fault behaves — whether it appears as a small leakage current that an RCD detects, or as a near-bolted-fault that triggers main breaker action.

Three schemes are common:

- **TN-S** (separate neutral and protective earth): first phase-to-earth fault is a full short circuit,

cleared by the breaker. Dominant in IT/PDU layers because it gives predictable short-circuit currents.
- **TN-C-S**: combined upstream, separated downstream; good cost-performance tradeoff but vulnerable to PEN-conductor breaks.
- **IT** (isolated neutral): first earth fault gives only a small displacement current; the second fault, on a different phase, becomes a phase-to-phase short. Used in some EU sites for continuity-of-service but requires insulation monitoring.

The choice affects how the elementary events $E_\bullet$ in the fault tree (4) compose: in an IT scheme, the first-fault event is harmless but the second-fault event is large; the joint distribution is therefore highly non-Markovian and an additional state must be added to the chain of Figure 3.

A.10. Putting the physics back into the hazard

We can now interpret each term of the hazard rate (12) in physical units:

$$\lambda_t \;=\; \lambda_0\, e^{-E_a/k_B T_t} \cdot \big(V_t/V_{\text{nom}}\big)^{-\gamma_V} \cdot \exp(\beta_C C_t).$$

- λ_0 has dimensions of events/year, and is obtained from the steady-state Markov-chain analysis of §4.2 together with the Thévenin/short-circuit analysis of §A.2.
- E_a is the AIPL activation energy of the insulation system (transformer paper, capacitor dielectrics), 0.6–0.9 eV.
- γ_V is the inverse-power-law voltage stress exponent from (73), calibrated from THD measurements (72) and dielectric endurance tests.
- $\beta_C C_t$ is the cyber-load Esscher tilt, with C_t in standardised log-rate units.

The deterministic grace period (17) contains UA and mc_p, which an EE/HVAC engineer measures directly during commissioning. *This is the closure of the loop*: every parameter in the actuarial pricing equation has a physical measurement attached, and the parameters are identified, not assumed.

References

[1] A. A. Balkema and L. de Haan, "Residual life time at great age," *Ann. Probab.*, vol. 2, pp. 792–804, 1974.

[2] J. R. Black, "Electromigration — a brief survey and some recent results," *IEEE Trans. Electron Dev.*, vol. ED-16, pp. 338–347, 1969.

[3] The Baldwin Group, "Data centers: risk and insurance across the development lifecycle," January 2026.

[4] R. S. Chhikara and J. L. Folks, *The Inverse Gaussian Distribution: Theory, Methodology, and Applications*, Marcel Dekker, 1989.

[5] CyberCube, "AWS US-East-1 Security Incident Report," preliminary loss range $38–581 M, October 2025.

[6] P. Sisson, "Report: Billions of dollars in data center construction risk is uninsured," *Engineering News-Record*, 2025.

[7] Uptime Institute Intelligence, "South Korean data center fire sparks a stark reminder," November 2025.

[8] A. Bednarz, "Data center fires raise concerns about lithium-ion batteries," *NetworkWorld*, March 2023.

[9] M. Lebovits, "Insuring the cloud," American Academy of Actuaries, September 2022.

[10] J. Pickands, "Statistical inference using extreme order statistics," *Ann. Statist.*, vol. 3, pp. 119–131, 1975.

[11] K. S. Trivedi, *Probability and Statistics with Reliability, Queuing and Computer Science Applications*, 2nd ed., Wiley, 2017.

[12] Uptime Institute, "Annual Outage Analysis 2024."

[13] S. S. Wang, "A class of distortion operators for pricing financial and insurance risks," *J. Risk Insur.*, vol. 67, pp. 15–36, 2000.